



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Data Analytics with Python

Lecture 1: Introduction to data analytics

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IIT ROORKEE



Objective of the course

- The principle focus of this course is to introduce conceptual understanding using simple and practical examples rather than repetitive and point click mentality
- This course should make you comfortable using analytics in your career and your life
- You will know how to work with real data, and might have learned many different methodologies but choosing the right methodology is important

Objective of the course Contd...

- The danger in using quantitative method does not generally lie in the inability to perform the calculation
- The real threat is lack of fundamental understanding of:
 - Why to use a particular technique or procedure
 - How to use it correctly and,
 - How to correctly interpret the result



Learning objectives

1. Define data and its importance
2. Define data analytics and its types
3. Explain why analytics is important in today's business environment
4. Explain how statistics, analytics and data science are interrelated
5. Why python?
6. Explain the four different levels of Data:
 - Nominal
 - Ordinal
 - Interval and
 - Ratio

1. Define Data and its importance

- Variable, Measurement and Data
- What is generating so much data?
- How data add value to the business?
- Why data is important?

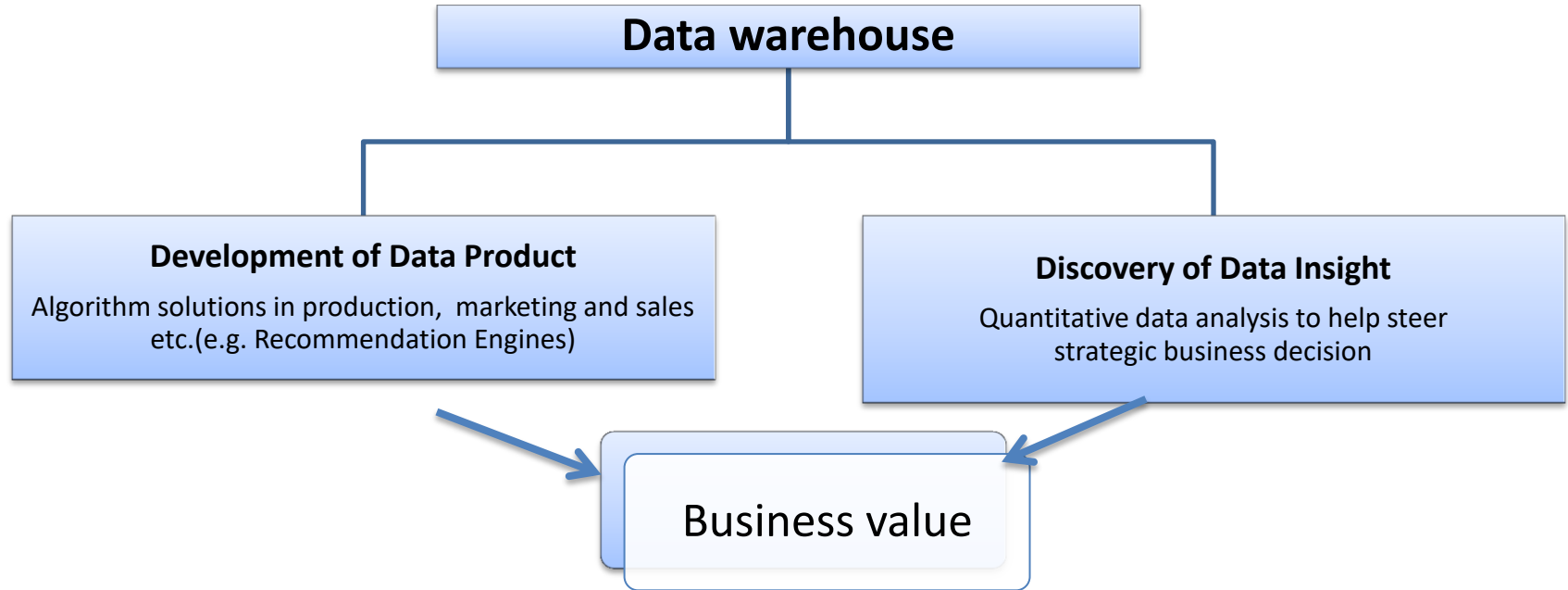
1.1 Variable, Measurement and Data

- Variables – is a characteristic of any entity being studied that is capable of taking on different values
- Measurements – is when a standard process is used to assign numbers to particular attributes or characteristic of a variable
- Data – data are recorded measurements

1.2 What is generating so much data?

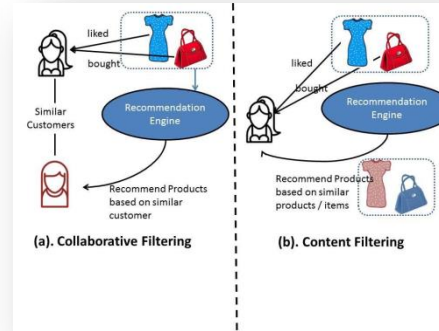
- Data can be generated by
 - Humans,
 - Machines or
 - Humans-machines combines
- It can be generated anywhere where any information is generated and stored in structured or unstructured formats

1.3 How data add value to business?



Source: <https://datajobs.com/>

Data Products



1.4 Why Data is important?

- Data helps in make better decisions
- Data helps in solve problems by finding the reason for underperformance
- Data helps one to evaluate the performance.
- Data helps one improve processes
- Data helps one understand consumers and the market

2. Define data analytic and its types

- Define data analytics
- Why analytics is important?
- Data analysis
- Data analytics vs. Data analysis
- Types of Data analytics

2.1. Define data analytics

- Analytics is defined as “the scientific process of transforming data into insights for making better decisions”
- Analytics, is the use of data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help managers gain improved insight about their business operations and make better, fact-based decisions – James Evans
- Analysis = Analytics ?

2.2 Why analytics is important?

- Opportunity abounds for the use of analytics and big data such as:



1. Determining credit risk
2. Developing new medicines
3. Finding more efficient ways to deliver products and services
4. Preventing fraud
5. Uncovering cyber threats
6. Retaining the most valuable customers



2.3 Data analysis

- Data analysis is the process of examining, transforming, and arranging raw data in a specific way to generate useful information from it
- Data analysis allows for the evaluation of data through analytical and logical reasoning to lead to some sort of outcome or conclusion in some context
- Data analysis is a multi-faceted process that involves a number of steps, approaches, and diverse techniques

Analysis

2.4 Data analytics vs. Data analysis

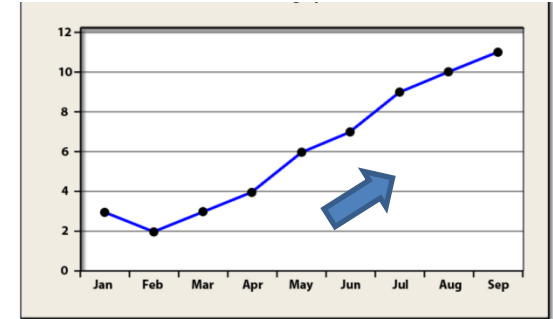


Past

Explain

How?

Why?



2.4 Data analytics vs. Data analysis

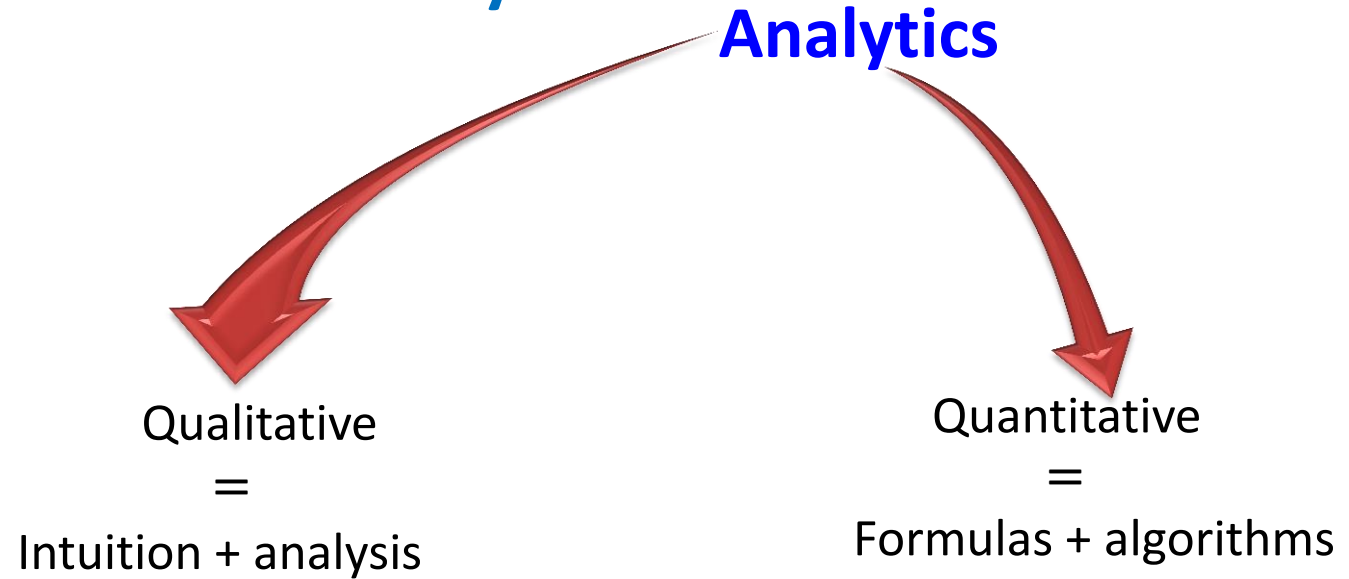
Analytics



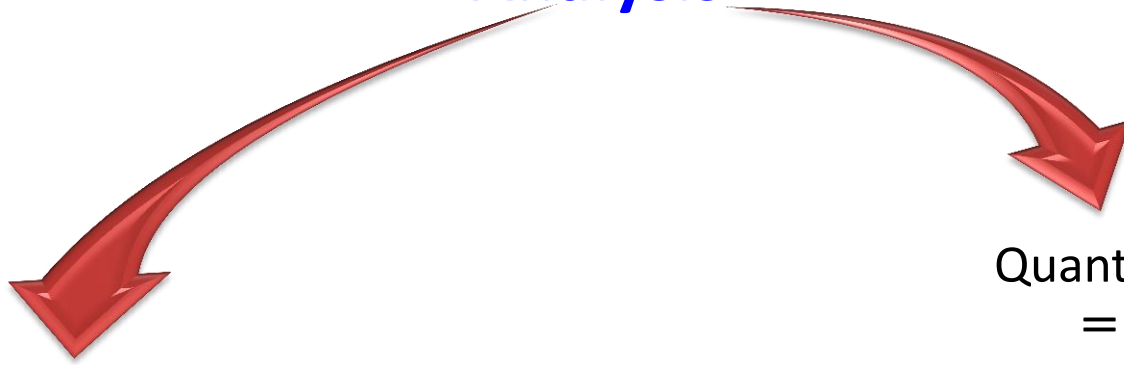
Future

Explore potential future events

2.4 Data analytics vs. Data analysis



Analysis



Qualitative

=

Explains **How** And **Why** Story ends the way it did ?

Quantitative

=

Data + **how** the sale **decreased** last summer

Analysis $\neq$ Analytics

Data Analysis $\neq$ Data analytics

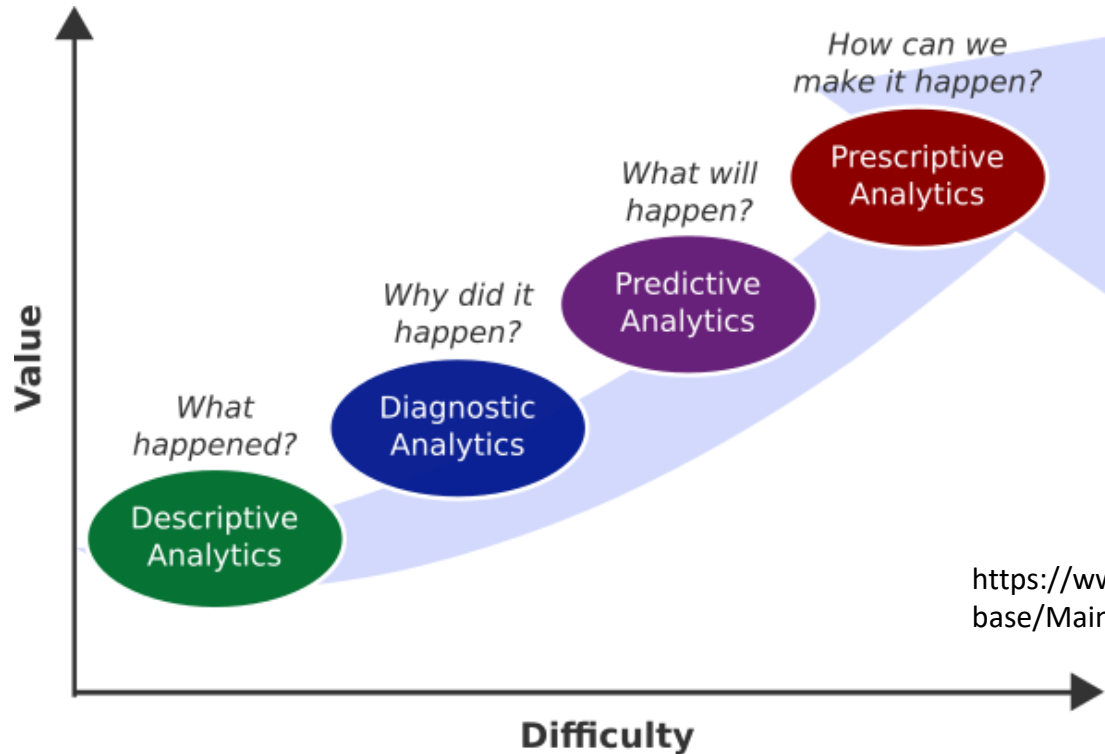
Business Analysis $\neq$ Business analytics

2.5 Classification of Data analytics

Based on the phase of workflow and the kind of analysis required, there are four major types of data analytics.

- Descriptive analytics
- Diagnostic analytics
- Predictive analytics
- Prescriptive analytics

Classification of Data analytics



https://www.governanceanalytics.org/knowledge-base/Main_Tools/Data_classification_and_analysis

Descriptive Analytics

- Descriptive Analytics, is the conventional form of Business Intelligence and data analysis
- It seeks to provide a depiction or “summary view” of facts and figures in an understandable format
- This either inform or prepare data for further analysis
- Descriptive analysis or statistics can summarize raw data and convert it into a form that can be easily understood by humans
- They can describe in detail about an event that has occurred in the past

Example

A common example of Descriptive Analytics are company reports that simply provide a historic review like:

- Data Queries
- Reports
- Descriptive Statistics
- Data Visualization
- Data dashboard



Source: <https://www.linkedin.com/learning/478e9692-d13d-338f-907e-d76f0724d773>

Diagnostic analytics

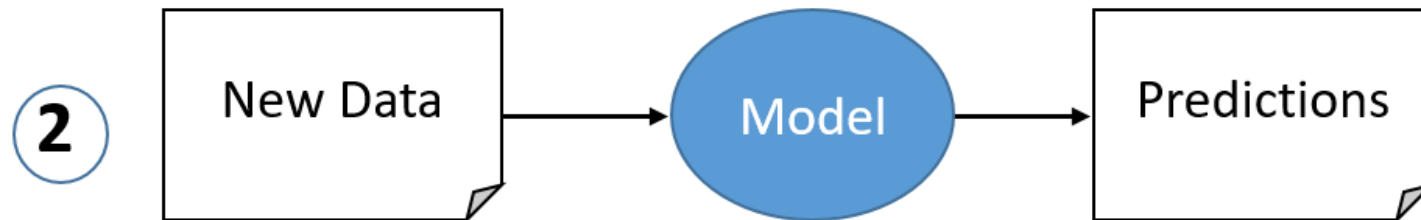
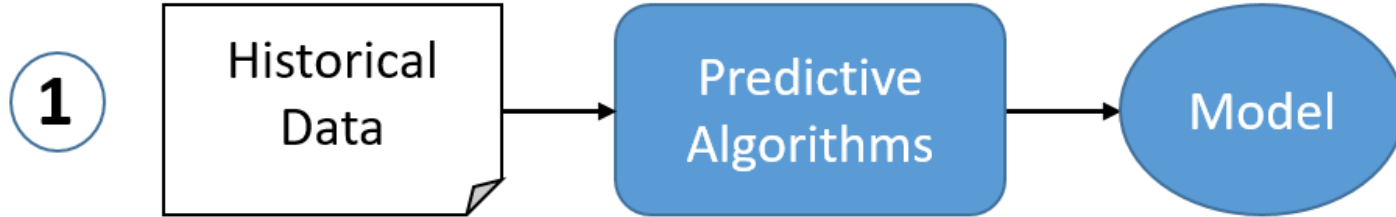
- Diagnostic Analytics is a form of advanced analytics which examines data or content to answer the question “Why did it happen?”
- Diagnostic analytical tools aid an analyst to dig deeper into an issue so that they can arrive at the source of a problem
- In a structured business environment, tools for both descriptive and diagnostic analytics go parallel

Example

- It uses techniques such as:
 1. Data Discovery
 2. Data Mining
 3. Correlations

Predictive analytics

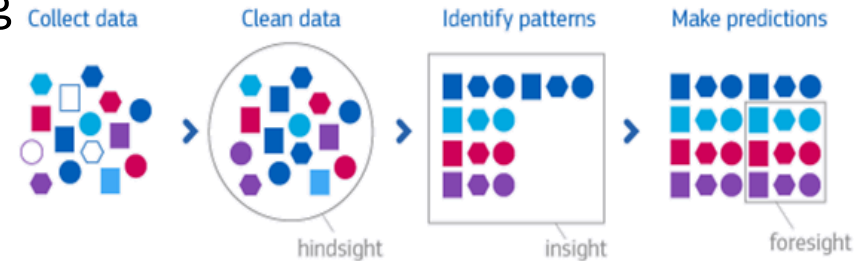
- Predictive analytics helps to forecast trends based on the current events
- Predicting the probability of an event happening in future or estimating the accurate time it will happen can all be determined with the help of predictive analytical models
- Many different but co-dependent variables are analysed to predict a trend in this type of analysis



Source: <https://www.logianalytics.com/wp-content/uploads/2017/11/predictive-1.png>

Example

- Set of techniques that use model constructed from past data to predict the future or ascertain impact of one variable on another:
 - Linear regression
 - Time series analysis and forecasting
 - Data mining



Source: <https://bigdata-madesimple.com/5-examples-predictive-analytics-travel-industry/>

Prescriptive analytics

- Set of techniques to indicate the best course of action
- It tells what decision to make to optimize the outcome
- The goal of prescriptive analytics is to enable:
 1. Quality improvements
 2. Service enhancements
 3. Cost reductions and
 4. Increasing productivity

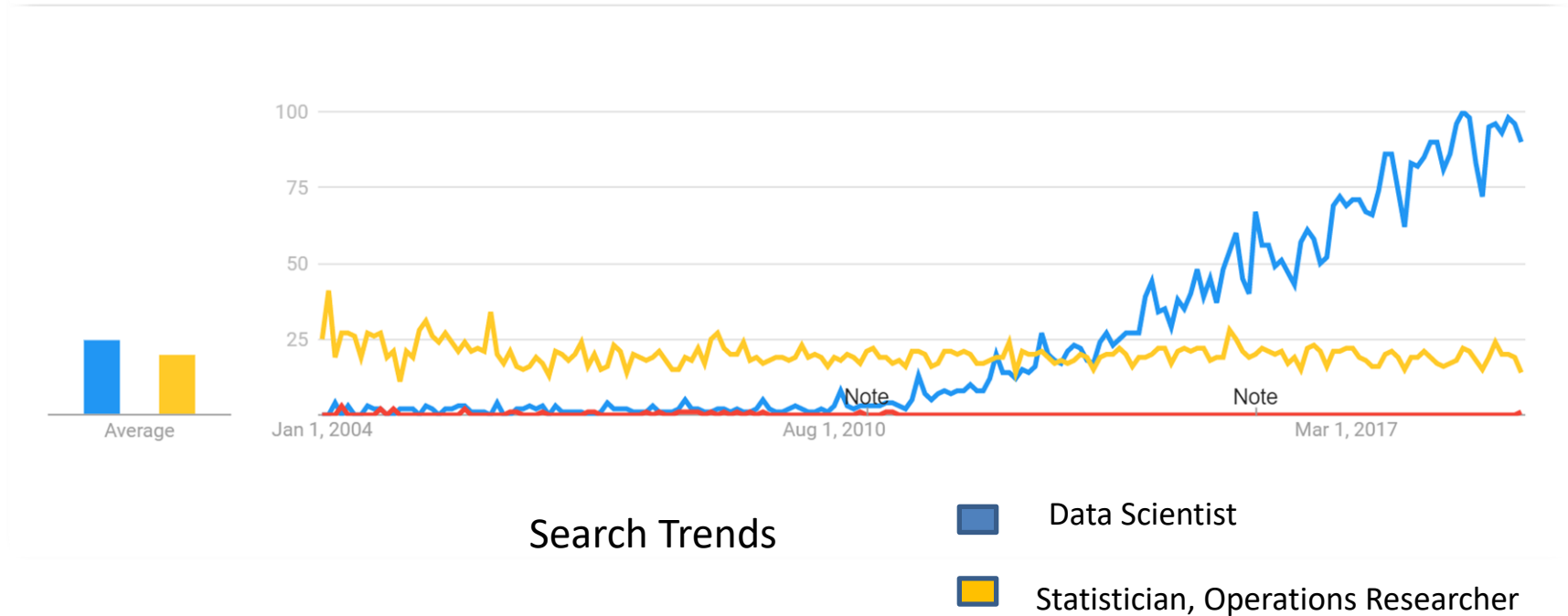
Prescriptive analytics: Example

- Optimization Model
- Simulation
- Decision Analysis

3. Explain why analytics is important

- Demand for Data Analytics
- Element of data Analytics

3. Explain why analytics is important





THIS STORY IS FROM MAY 8, 2016

Data scientists earning more than CAs, engineers

Namrata Singh | TNN | Updated: May 8, 2016, 14:50 IST



A-

A+

<https://timesofindia.indiatimes.com/india/Data-scientists-earning-more-than-CAs-engineers/articleshow/52171064.cms>

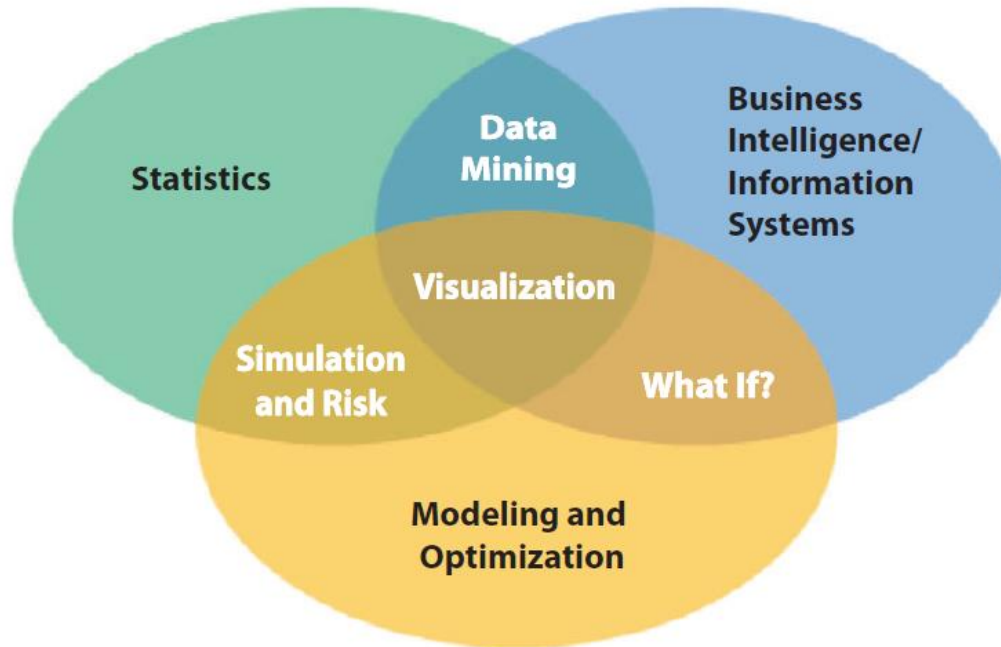
3.1 Demand for Data Analytics

With companies across industries striving to bring their research and analysis (R&A) departments up to speed, the demand for qualified data scientists is rising.

"India will face a demand-supply gap of 2,00,000 analytics professionals over the next three years. Even in the US, only 40 out of 100 positions for analytics professionals can be filled," said Rituparna Chakraborty , co-founder & senior VP of TeamLease Services.

http://timesofindia.indiatimes.com/articleshow/52171064.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst

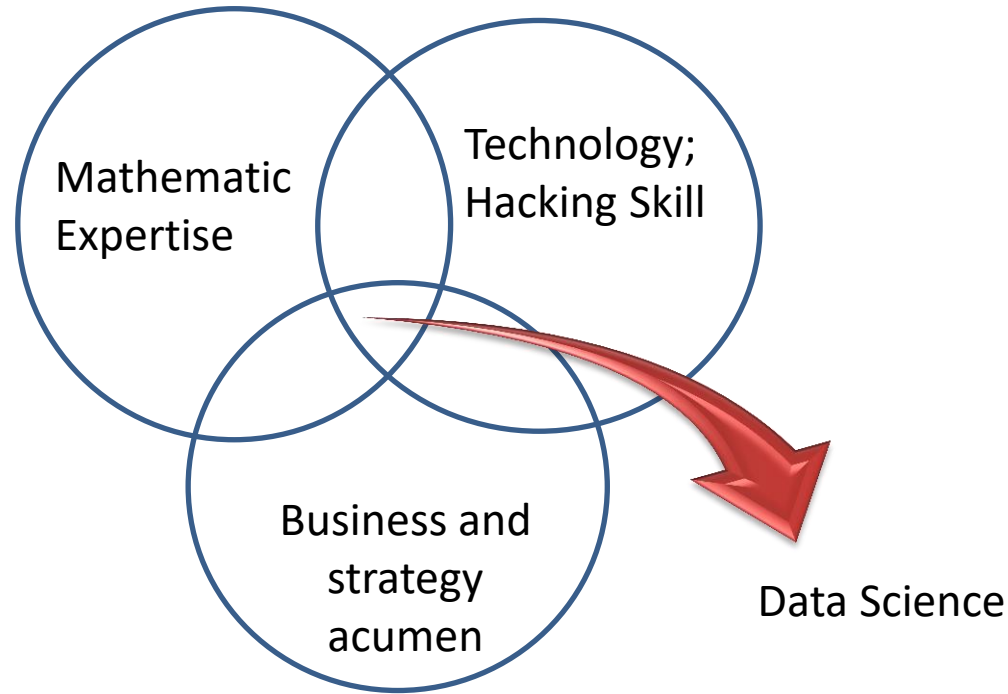
3.2 Element of data Analytics



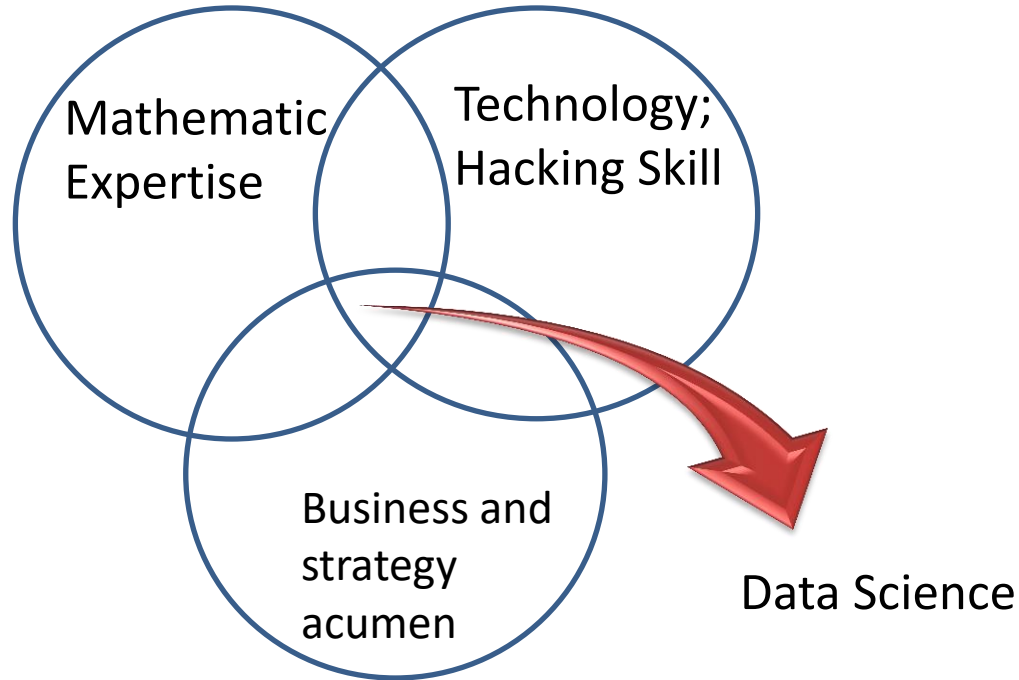
4. Data analyst and Data scientist

- The requisite skill set
- Difference between Data analyst and Data Scientist

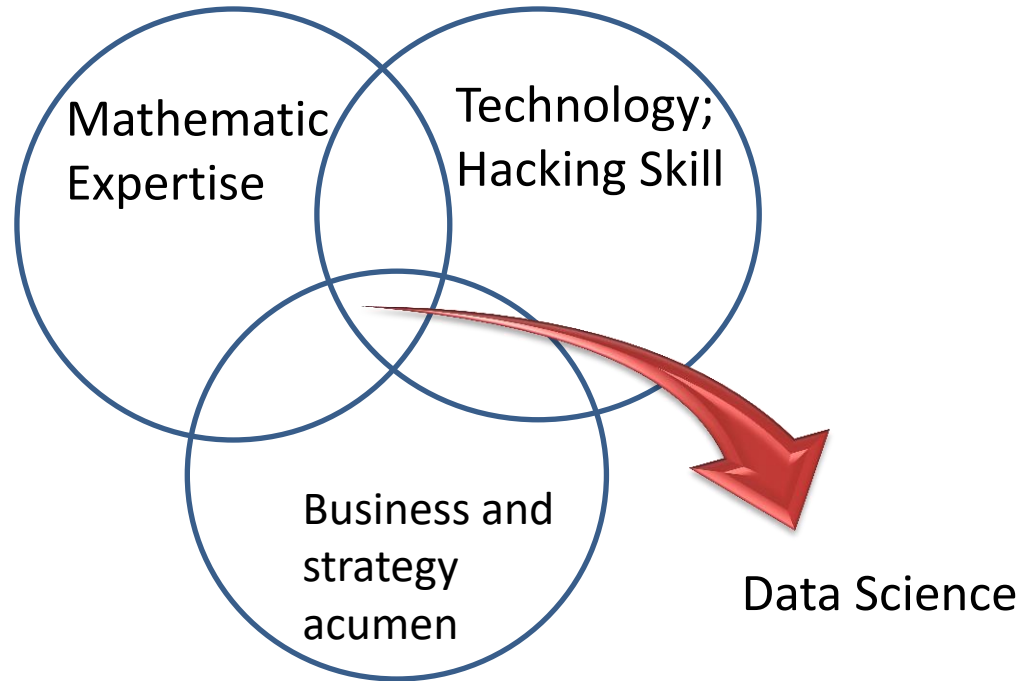
4.1 The requisite skill set



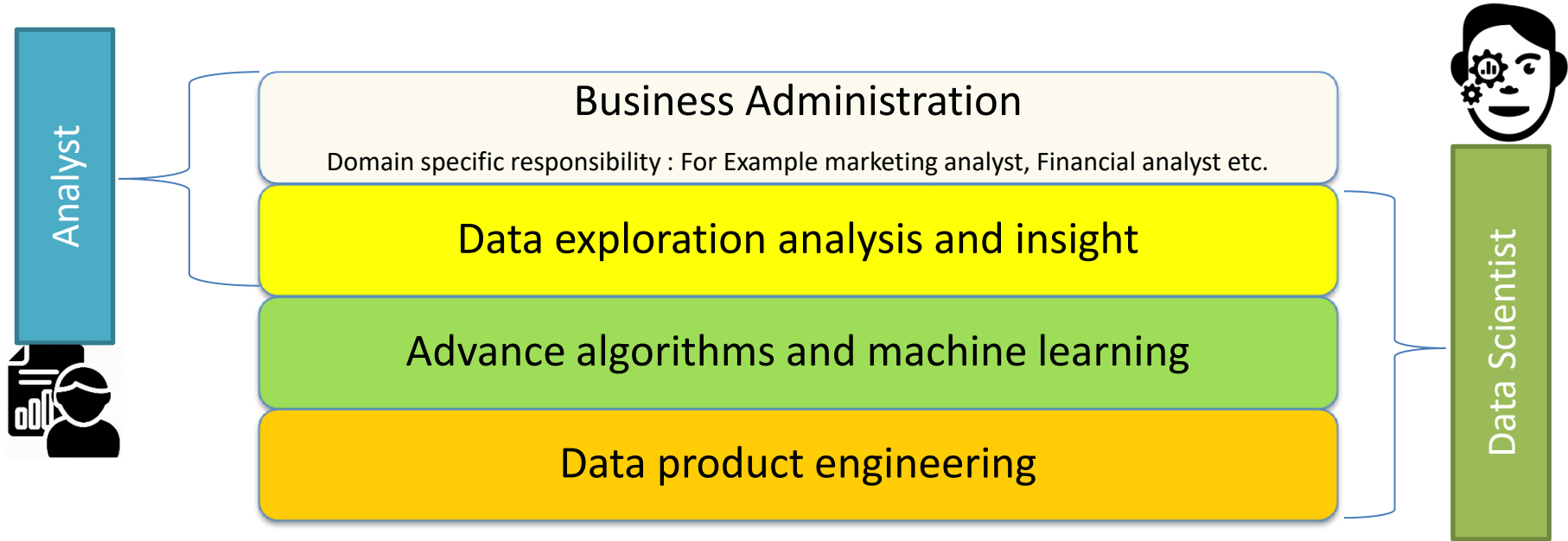
4.1 The requisite skill set



4.1 The requisite skill set



4.2 Difference between Data analyst and Data Scientist



Source: <https://datajobs.com/>

5. Why python?



Features

- Simple and easy to learn
- Freeware and Open source
- Interpreted
- Dynamically Typed
- Extensible
- Embedded
- Extensive library

5. Why python?



Usability

- Desktop and web applications
- Database applications
- Networking applications
- Data analysis (Data Science)
- Machine learning
- IoT and AI applications
- Games

Companies using Python



Why Jupyter Notebook?



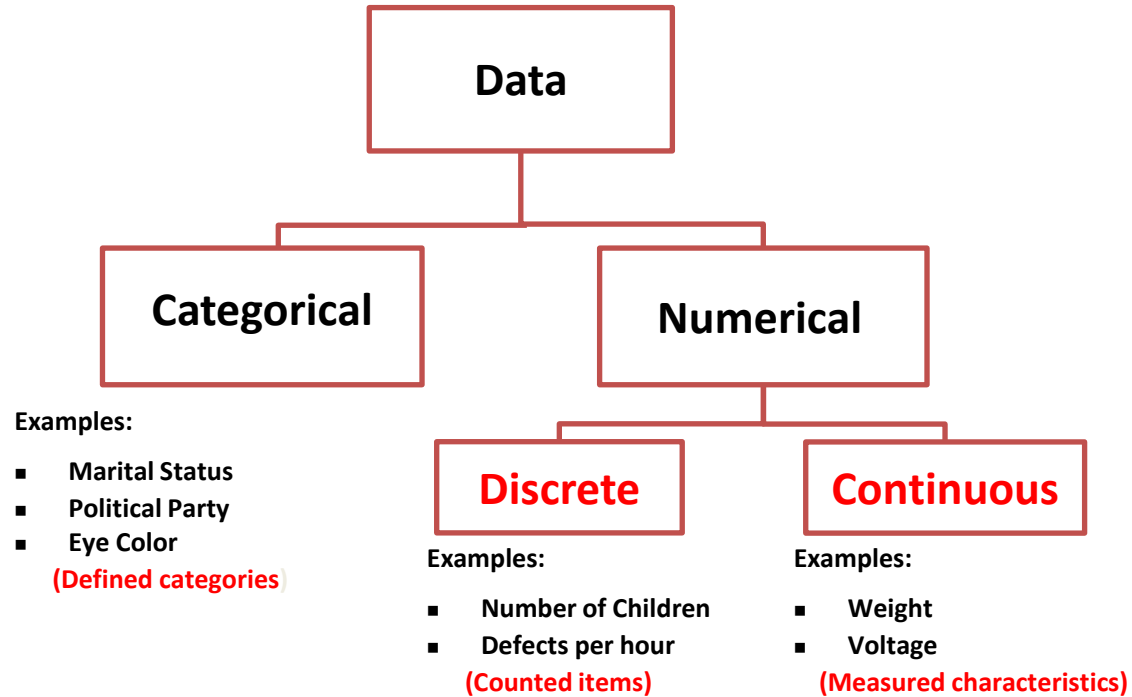
Why?

- Client – Server Application
- Edit code on web browser
- Easy in documentation
- Easy in demonstration
- User- friendly Interface

6. Explain the four different levels of Data

- Types of Variables
- Levels of Data Measurement
- Compare the four different levels of Data:
Nominal
Ordinal
Interval and
Ratio
- Usage Potential of Various Levels of Data
- Data Level, Operations, and Statistical Methods

6.1 Types of Variables



6.2 Levels of Data Measurement

- Nominal — Lowest level of measurement
- Ordinal
- Interval
- Ratio — Highest level of measurement

6.3.1 Nominal

- A **nominal scale** classifies data into distinct categories in which no ranking is implied
- Example : Gender, Marital Status

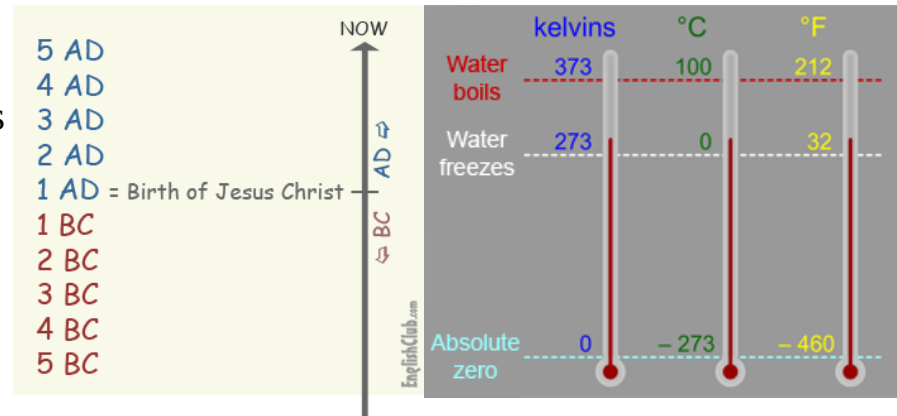


6.3.2 Ordinal scale

- An **ordinal scale** classifies data into distinct categories in which ranking is implied
- Example:
 - Product satisfaction → Satisfied, Neutral, Unsatisfied
 - Faculty rank → Professor, Associate Professor, Assistant Professor
 - Student Grades → A, B, C, D, F

6.3.3. Interval scale

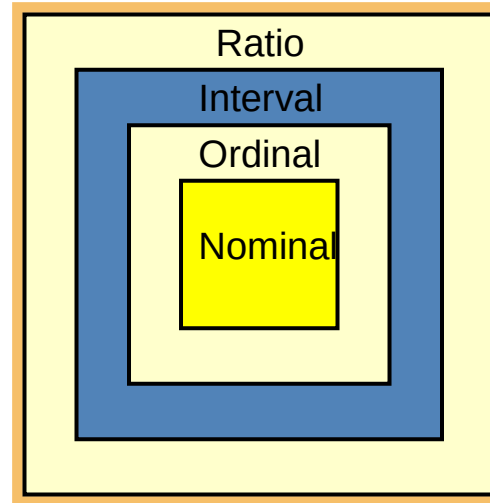
- An **interval scale** is an ordered scale in which the difference between measurements is a meaningful quantity but the measurements do not have a true zero point.
- Example
 - Temperature in Fahrenheit and Celsius
 - Year



6.3.4 Ratio scale

- A **ratio scale** is an ordered scale in which the difference between the measurements is a meaningful quantity and the measurements have a true zero point.
- Example
 - Weight
 - Age
 - Salary

6.4 Usage Potential of Various Levels of Data



6.5 Impact of choice of measurement scale

Data Level	Meaningful Operations	Statistical Methods
Nominal	Classifying and Counting	Nonparametric
Ordinal	All of the above plus Ranking	Nonparametric
Interval	All of the above plus Addition, Subtraction	Parametric
Ratio	All of the above plus multiplication and division	Parametric

Thank You





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Data Analytics with Python

Lecture 2: Python – Fundamentals

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IIT ROORKEE



Learning objectives

1. Installing Python
2. Fundamentals of Python
3. Data Visualisation

Python Installation Process

Installation Process –

Step 1: Type <https://www.anaconda.com> at the address bar of web browser.

Step 2: Click on download button

Step 3: Download python 3.7 version for windows OS

Step 4: Double click on file to run the application

Step 5: Follow the instructions until completion of installation process

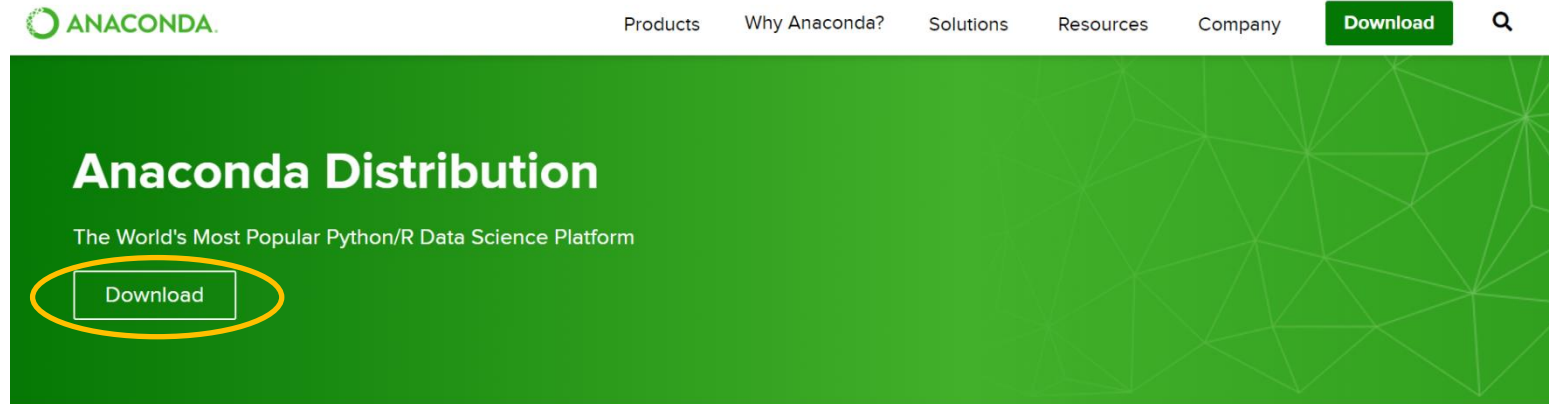
Python Installation Process

Installation Process –

Step 1: Type <https://www.anaconda.com> at the address bar of web browser.

Python Installation Process

Step 2: Click on download button



ANACONDA.

Products Why Anaconda? Solutions Resources Company **Download** 🔍

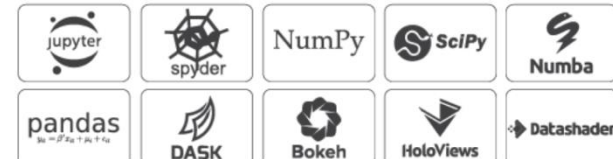
Anaconda Distribution

The World's Most Popular Python/R Data Science Platform

Download

The open-source **Anaconda Distribution** is the easiest way to perform Python/R data science and machine learning on Linux, Windows, and Mac OS X. With over 11 million users worldwide, it is the industry standard for developing, testing, and training on a single machine, enabling *individual data scientists* to:

- Quickly download 1,500+ Python/R data science packages



Python Installation Process

Step 3: Download python 3.7 version for windows OS

 Windows |  macOS |  Linux

Anaconda 2019.03 for Windows Installer

Python 3.7 version

Download

64-Bit Graphical Installer (662 MB)
32-Bit Graphical Installer (546 MB)

Python 2.7 version

Download

64-Bit Graphical Installer (587 MB)
32-Bit Graphical Installer (493 MB)

Get Started with Anaconda Distribution

Python Installation Process

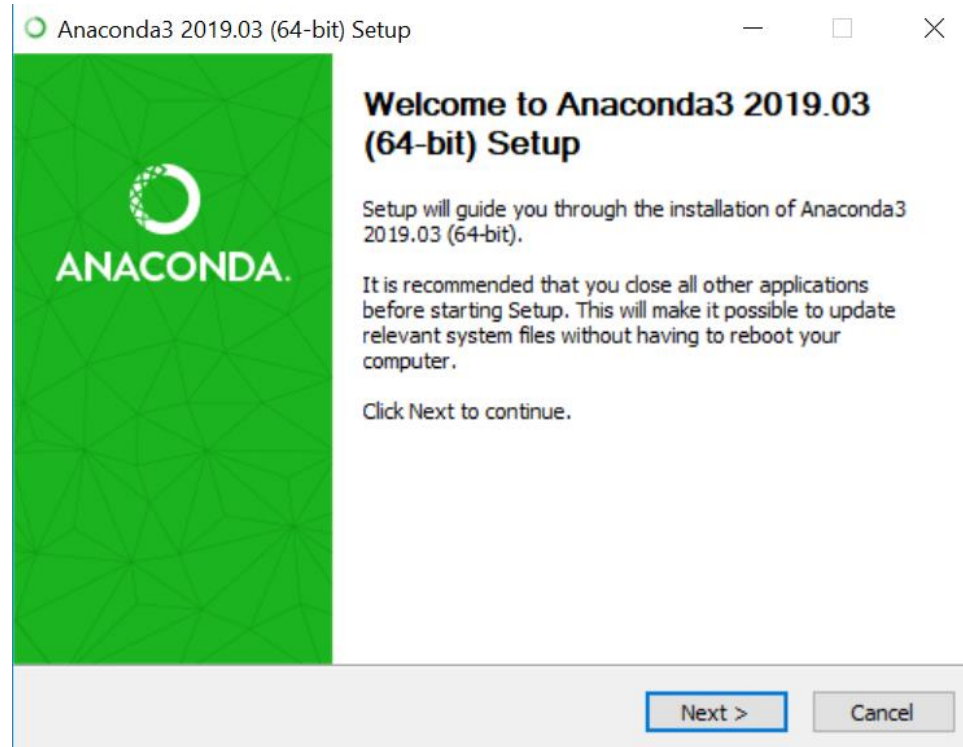
Step 4: Double click on file to run the application



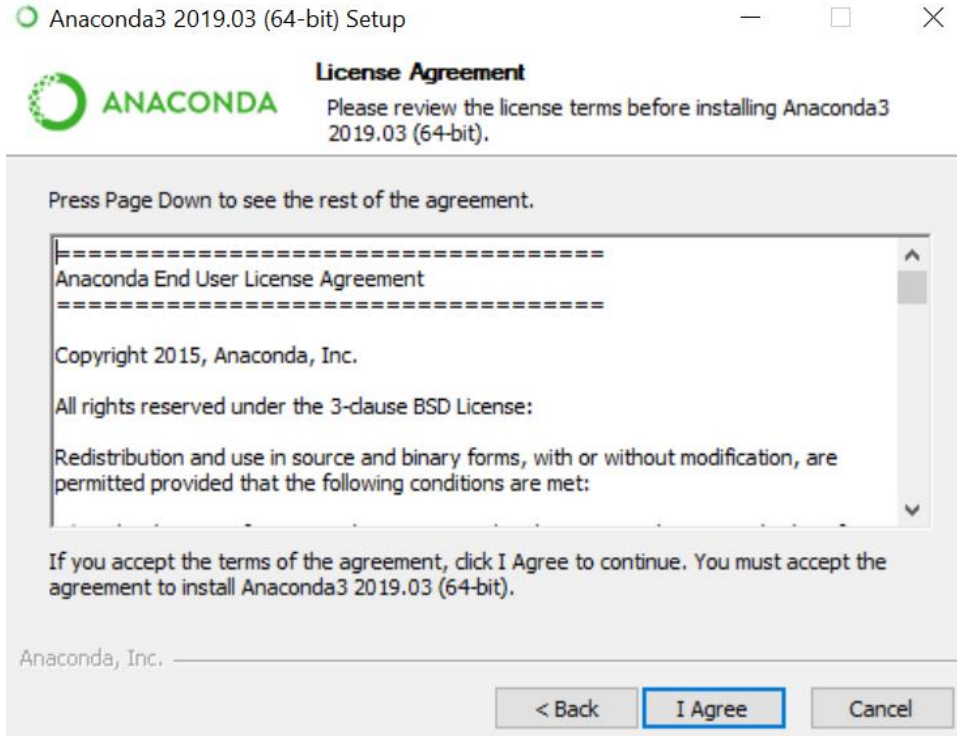
Anaconda3-2018.12-Windows-x86_64_2.exe

F:\Umabharati

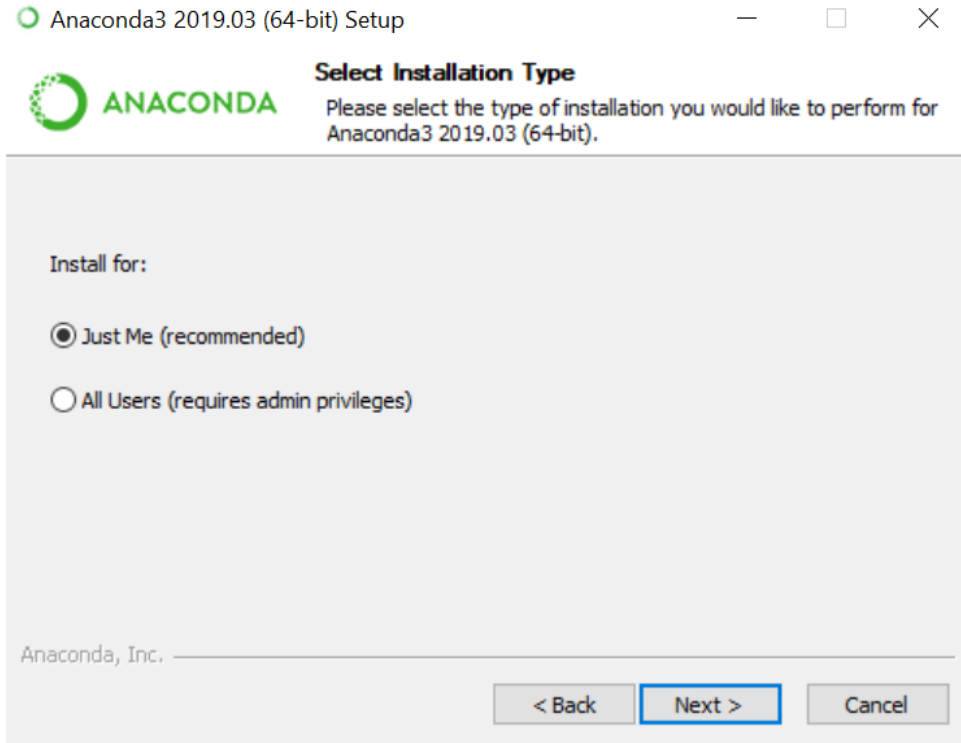
Python Installation Process



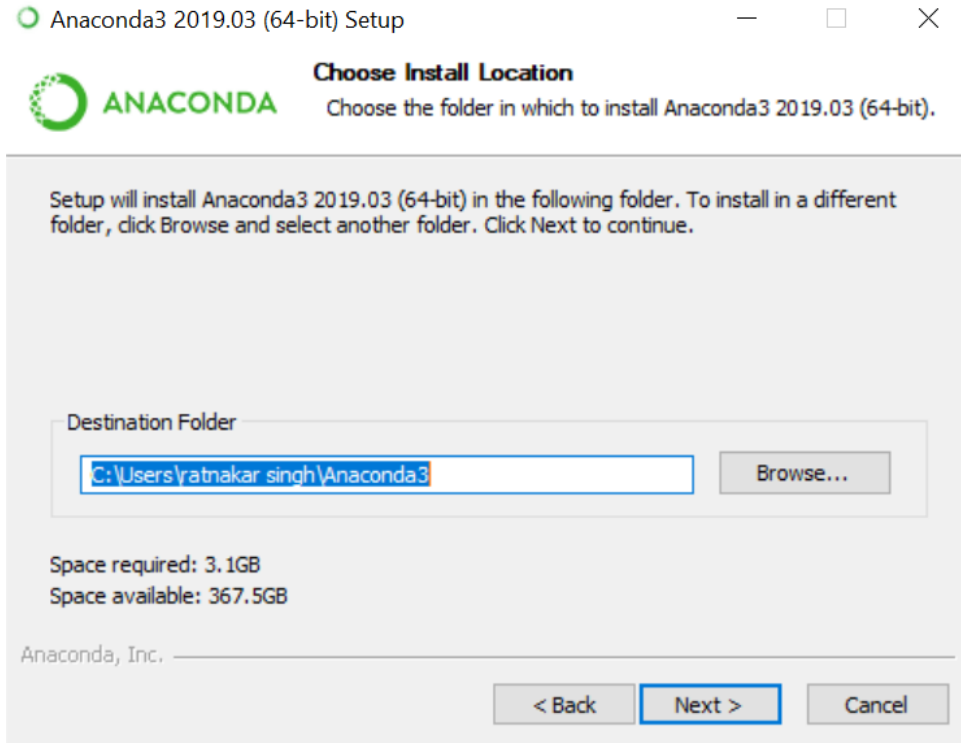
Python Installation Process



Python Installation Process



Python Installation Process



Python Installation Process

Anaconda3 2019.03 (64-bit) Setup



Advanced Installation Options

Customize how Anaconda integrates with Windows

Advanced Options

☒ Add Anaconda to my PATH environment variable

Not recommended. Instead, open Anaconda with the Windows Start menu and select "Anaconda (64-bit)". This "add to PATH" option makes Anaconda get found before previously installed software, but may cause problems requiring you to uninstall and reinstall Anaconda.

☒ Register Anaconda as my default Python 3.7

This will allow other programs, such as Python Tools for Visual Studio, PyCharm, Wing IDE, PyDev, and MSI binary packages, to automatically detect Anaconda as the primary Python 3.7 on the system.

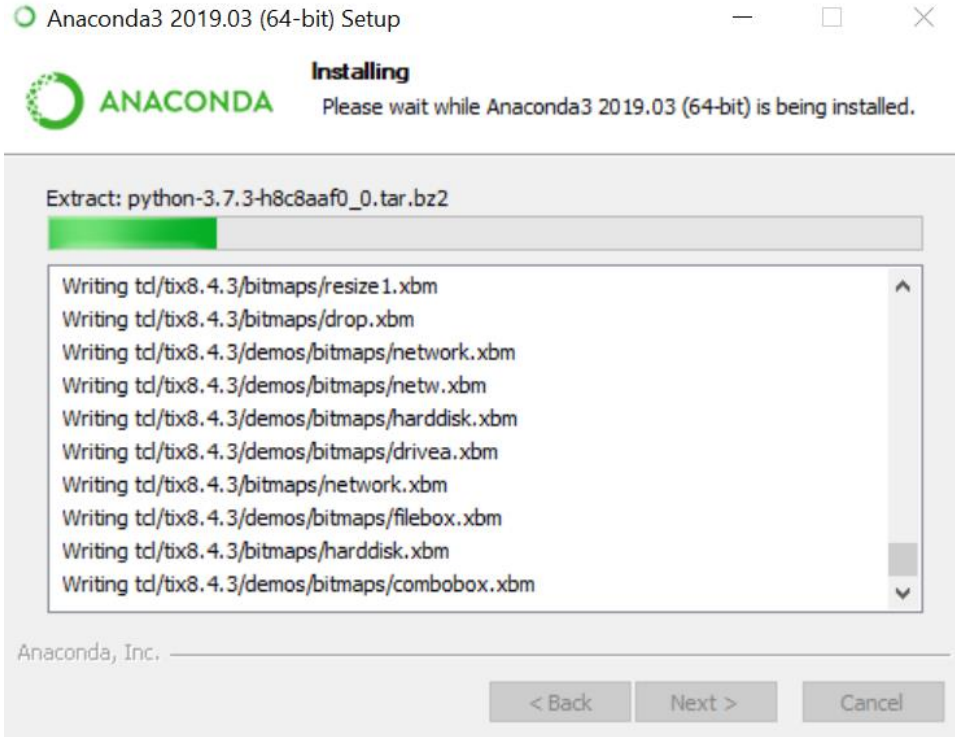
Anaconda, Inc.

< Back

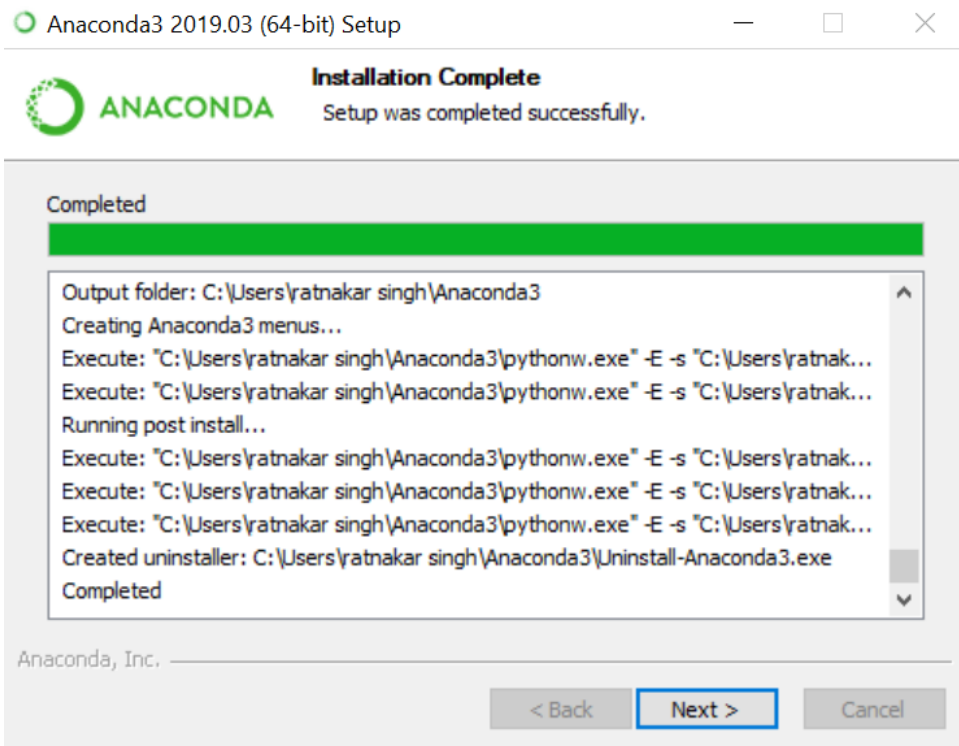
Install

Cancel

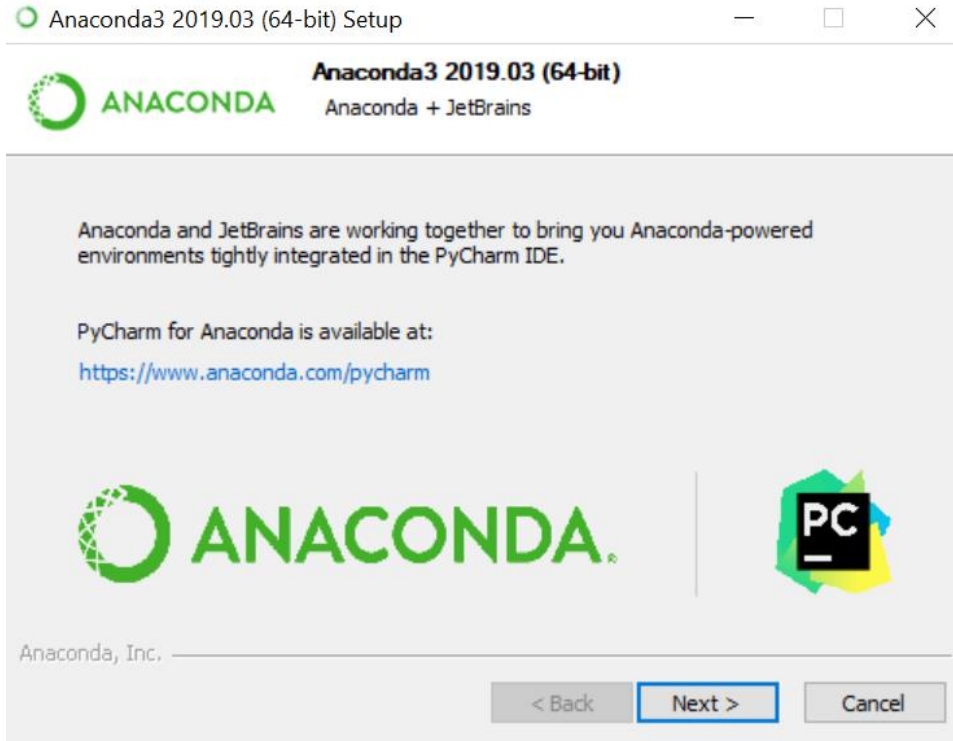
Python Installation Process



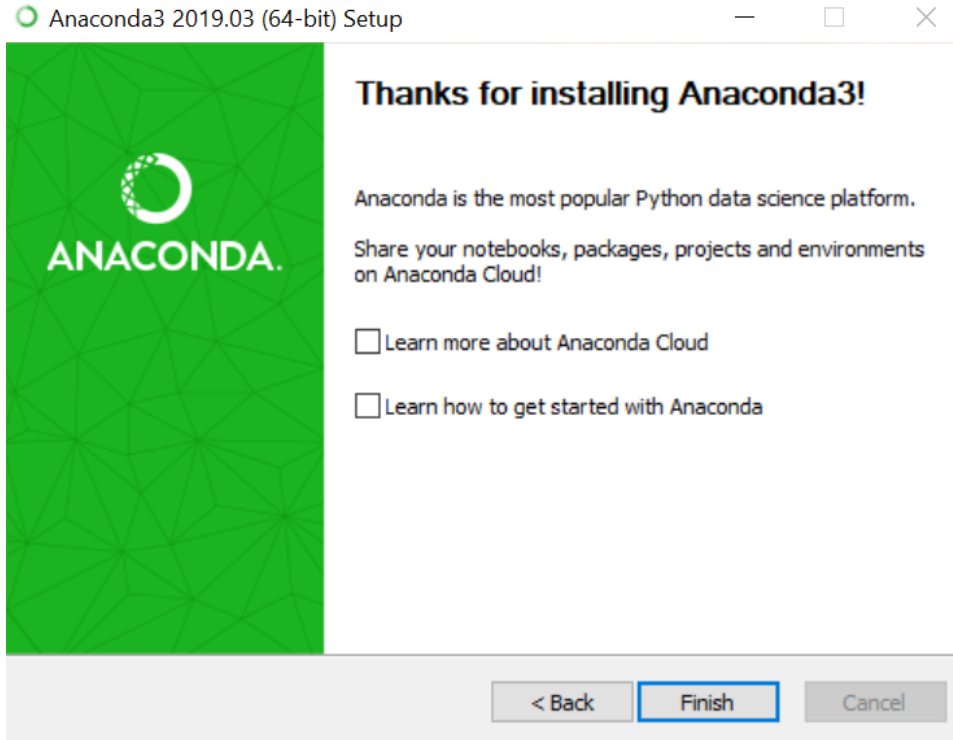
Python Installation Process



Python Installation Process



Python Installation Process



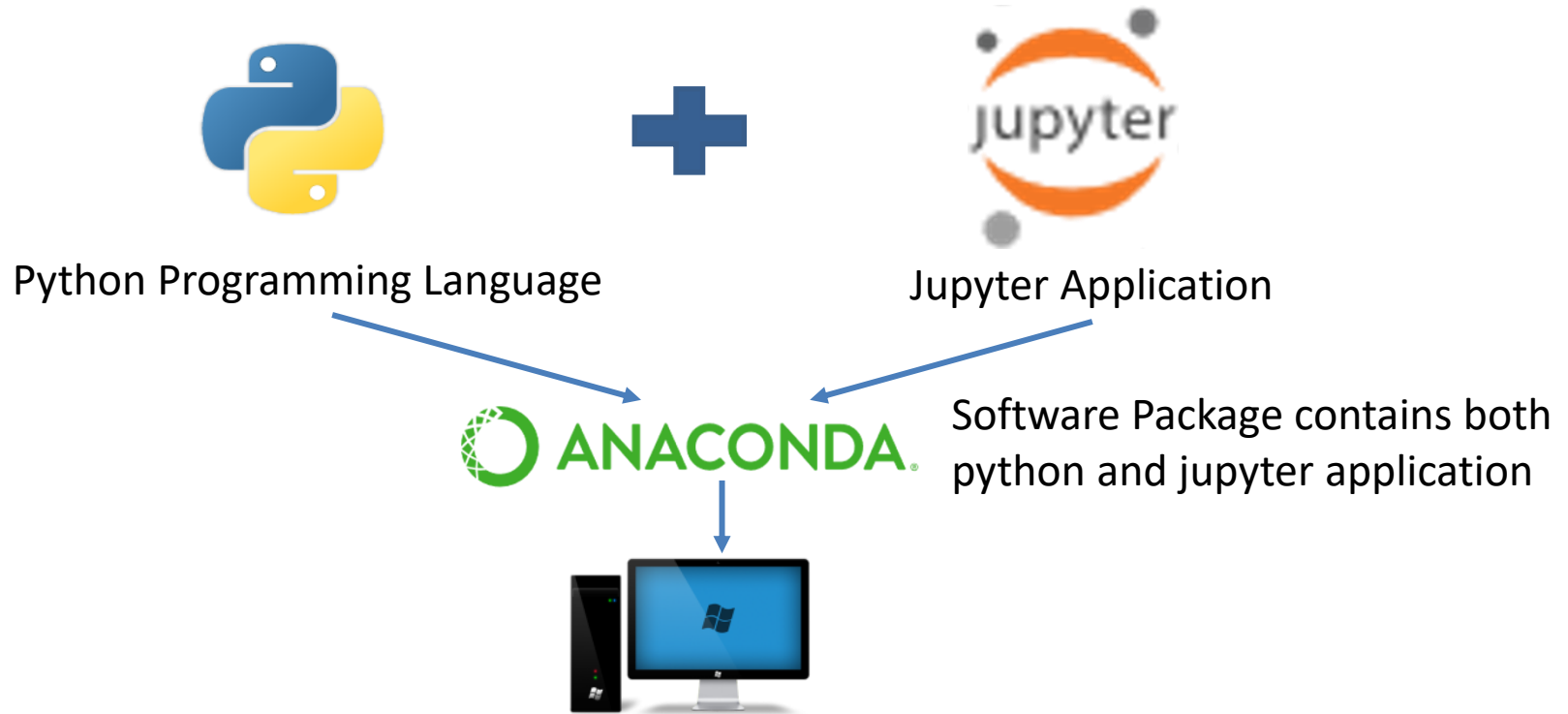
Why Jupyter Notebook?



Why?

- Edit code on web browser
- Easy in documentation
- Easy in demonstration
- User- friendly Interface

Python and Jupyter



Applications on base (root)

Channels

Refresh

 jupyterlab 0.31.4 An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture. Launch	 jupyter notebook 5.4.0 Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis. Launch	 qtconsole 4.3.1 PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more. Launch	 spyder 3.2.6 Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features Launch	 glueviz 0.12.0 Multidimensional data visualization across files. Explore relationships within and among related datasets. Install	 orange3 3.4.1 Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox. Install
 rstudio 1.1.383 A set of integrated tools designed to help you be more productive with R. Includes R essentials and notebooks. Install	 vscode 1.36.1 Streamlined code editor with support for development operations like debugging, task running and version control. Install				

About Jupyter Notebook

http://localhost:8888/notebooks/About_Jupyter_Notebook.ipynb

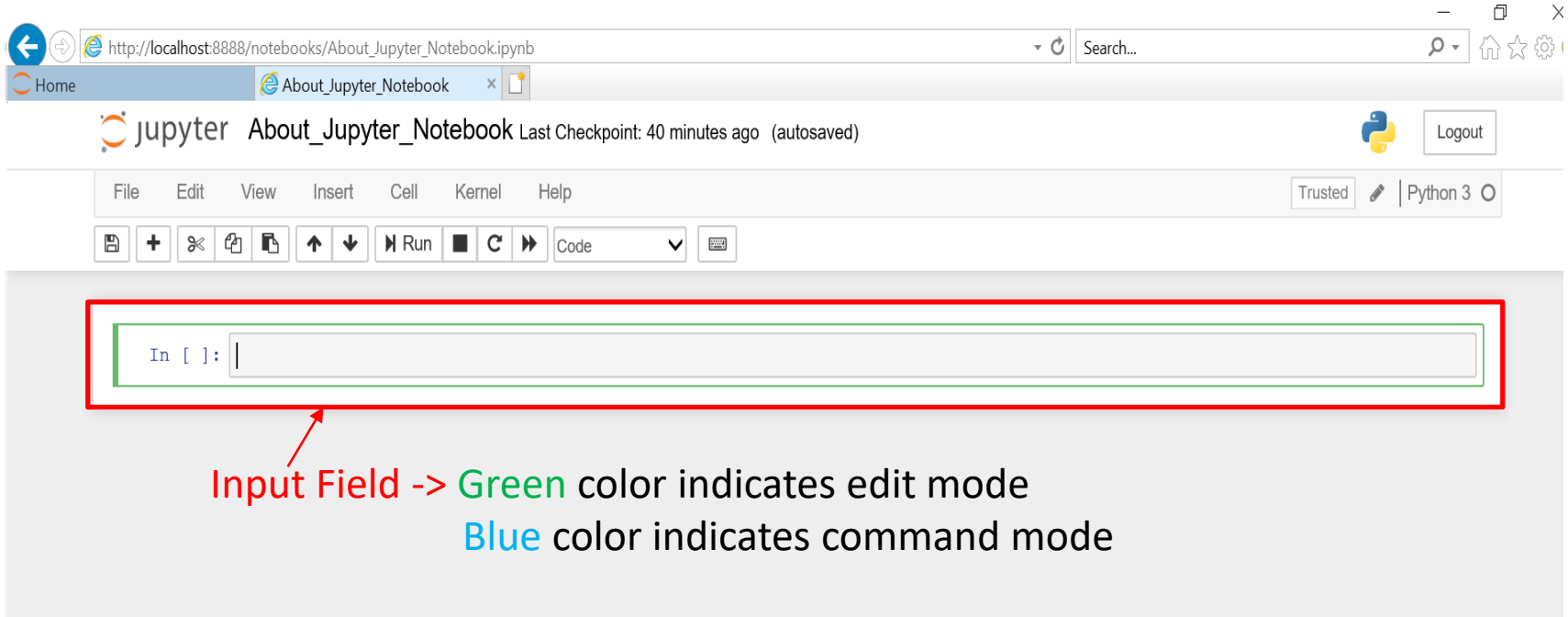
jupyter About_Jupyter_Notebook Last Checkpoint: a minute ago (unsaved changes)

File Edit View Insert Cell Kernel Help Trusted Python 3

In []:

Cell -> Access using Enter Key

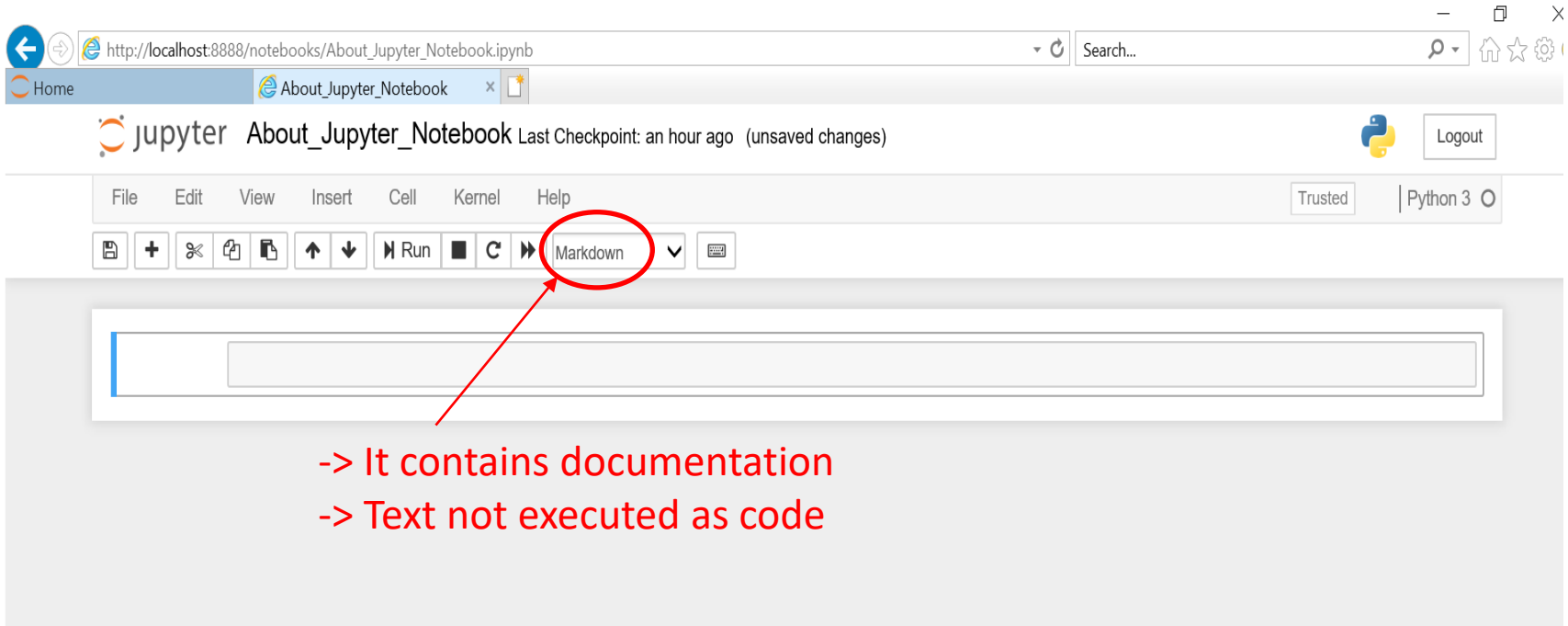
About Jupyter Notebook



The screenshot displays the Jupyter Notebook web interface in a browser. The address bar shows the URL `http://localhost:8888/notebooks/About_Jupyter_Notebook.ipynb`. The notebook title is "About_Jupyter_Notebook" with a status of "Last Checkpoint: 40 minutes ago (autosaved)". The top menu bar includes File, Edit, View, Insert, Cell, Kernel, and Help. Below the menu is a toolbar with icons for saving, adding, deleting, and running cells. The main area shows a code cell with the prompt `In [ ]:` followed by a cursor. The input field is highlighted with a red border, and a red arrow points to it from the text below. The text below the screenshot explains that the green color indicates edit mode and the blue color indicates command mode.

Input Field -> Green color indicates edit mode
Blue color indicates command mode

About Jupyter Notebook



The screenshot displays the Jupyter Notebook web interface in a browser. The address bar shows the URL `http://localhost:8888/notebooks/About_Jupyter_Notebook.ipynb`. The page title is "About_Jupyter_Notebook" with a subtitle "Last Checkpoint: an hour ago (unsaved changes)". The top navigation bar includes "File", "Edit", "View", "Insert", "Cell", "Kernel", and "Help". Below this is a toolbar with icons for saving, adding, deleting, and running cells. The "Markdown" button is circled in red, and a red arrow points to it from the text below. The main content area is a large text input field.

-> It contains documentation
-> Text not executed as code

About Jupyter Notebook

- Command mode allow to edit notebook as whole
- To close edit mode (Press Escape key)
- Execution (Three ways)
 - Ctrl +Enter (Output field can not be modified)
 - Shift +Enter (Output field is modified)
 - Run button on Jupyter interface
- Comment line is written preceding with # symbol.

About Jupyter Notebook

- Important shortcut keys
 - A -> To create cell above
 - B -> To create cell below
 - D + D -> For deleting cell
 - M -> For markdown cell
 - Y -> For code cell

Fundamentals of Python

- Loading a simple delimited data file
- Counting how many rows and columns were loaded
- Determining which type of data was loaded
- Looking at different parts of the data by subsetting rows and columns

Pandas for Everyone

Python Data Analysis

Daniel Y. Chen

◆◆Addison-Wesley

Boston • Columbus • Indianapolis • New York • San Francisco • Amsterdam • Cape Town

Dubai • London • Madrid • Milan • Munich • Paris • Montreal • Toronto • Delhi • Mexico City

São Paulo • Sydney • Hong Kong • Seoul • Singapore • Taipei • Tokyo



Loading a simple delimited data file

```
In [1]: import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt
```

```
In [23]: df = pandas.read_csv('F:/2019-20/NPTEL/2 Introduction to Python/data/gapminder-FiveYearData.csv')
```

```
In [20]: df
```

[Data Source: www.github.com/jennybc/gapminder.](https://www.github.com/jennybc/gapminder)

Out[29]:

	country	year	pop	continent	lifeExp	gdpPercap
0	Afghanistan	1952	8425333.0	Asia	28.801	779.445314
1	Afghanistan	1957	9240934.0	Asia	30.332	820.853030
2	Afghanistan	1962	10267083.0	Asia	31.997	853.100710
3	Afghanistan	1967	11537966.0	Asia	34.020	836.197138
4	Afghanistan	1972	13079460.0	Asia	36.088	739.981106
5	Afghanistan	1977	14880372.0	Asia	38.438	786.113360
6	Afghanistan	1982	12881816.0	Asia	39.854	978.011439
7	Afghanistan	1987	13867957.0	Asia	40.822	852.395945

- `head` method shows us only the first 5 rows

```
In [24]: print(df.head())
```

	country	year	pop	continent	lifeExp	gdpPercap
0	Afghanistan	1952	8425333.0	Asia	28.801	779.445314
1	Afghanistan	1957	9240934.0	Asia	30.332	820.853030
2	Afghanistan	1962	10267083.0	Asia	31.997	853.100710
3	Afghanistan	1967	11537966.0	Asia	34.020	836.197138
4	Afghanistan	1972	13079460.0	Asia	36.088	739.981106

Get the number of rows and columns

```
In [25]: print(df.shape)  
(1704, 6)
```

get column names

```
In [26]: print(df.columns)
```

```
Index(['country', 'year', 'pop', 'continent', 'lifeExp', 'gdpPercap'], dtype='object')
```

get the dtype of each column

```
In [27]: print(df.dtypes)
```

```
country      object  
year         int64  
pop          float64  
continent    object  
lifeExp      float64  
gdpPercap    float64  
dtype: object
```

Pandas Types Versus Python Types

Pandas Type	Python Type	Description
object	string	Most common data type
int64	int	Whole numbers
float64	float	Numbers with decimals
datetime64	datetime	datetime is found in the Python standard library (i.e., it is not loaded by default and needs to be imported)

get more information about data

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1704 entries, 0 to 1703  
Data columns (total 6 columns):  
country      1704 non-null object  
continent    1704 non-null object  
year         1704 non-null int64  
lifeExp      1704 non-null float64  
pop          1704 non-null int64  
gdpPercap    1704 non-null float64  
dtypes: float64(2), int64(2), object(2)  
memory usage: 80.0+ KB  
None
```

Looking at Columns, Rows, and Cells

- *# get the country column and save it to its own variable*

```
In [31]: country_df = df['country']
```

show the first 5 observations

```
In [32]: print(country_df.head())
```

```
0    Afghanistan
1    Afghanistan
2    Afghanistan
3    Afghanistan
4    Afghanistan
Name: country, dtype: object
```

show the last 5 observations

```
In [33]: print(country_df.tail())
```

```
1699    Zimbabwe  
1700    Zimbabwe  
1701    Zimbabwe  
1702    Zimbabwe  
1703    Zimbabwe  
Name: country, dtype: object
```

Looking at country, continent, and year

```
In [34]: subset = df[['country', 'continent', 'year']]  
print(subset.head())
```

	country	continent	year
0	Afghanistan	Asia	1952
1	Afghanistan	Asia	1957
2	Afghanistan	Asia	1962
3	Afghanistan	Asia	1967
4	Afghanistan	Asia	1972

```
In [35]: print(subset.tail())
```

	country	continent	year
1699	Zimbabwe	Africa	1987
1700	Zimbabwe	Africa	1992
1701	Zimbabwe	Africa	1997
1702	Zimbabwe	Africa	2002
1703	Zimbabwe	Africa	2007



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Data Analytics with Python

Lecture 3: Python – Fundamentals - II

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT
IIT ROORKEE



Looking at Columns, Rows, and Cells

- **Subset Rows by Index Label: loc**

In [36]: `print(df.head())`

	country	year	pop	continent	lifeExp	gdpPercap
0	Afghanistan	1952	8425333.0	Asia	28.801	779.445314
1	Afghanistan	1957	9240934.0	Asia	30.332	820.853030
2	Afghanistan	1962	10267083.0	Asia	31.997	853.100710
3	Afghanistan	1967	11537966.0	Asia	34.020	836.197138
4	Afghanistan	1972	13079460.0	Asia	36.088	739.981106

get the first row

- *Python counts from 0*

```
In [37]: print(df.loc[0])
```

country	Afghanistan
year	1952
pop	8.42533e+06
continent	Asia
lifeExp	28.801
gdpPercap	779.445
Name: 0, dtype: object	

- *# get the 100th row*
Python counts from 0

```
In [38]: print(df.loc[99])
```

```
country      Bangladesh  
year         1967  
pop          6.28219e+07  
continent     Asia  
lifeExp       43.453  
gdpPercap     721.186  
Name: 99, dtype: object
```

- *get the last row*

```
In [39]: print(df.tail(n=1))
```

	country	year	pop	continent	lifeExp	gdpPercap
1703	Zimbabwe	2007	12311143.0	Africa	43.487	469.709298

Subsetting Multiple Rows

- *# select the first, 100th, and 1000th rows*

```
In [40]: print(df.loc[[0, 99, 999]])
```

	country	year	pop	continent	lifeExp	gdpPercap
0	Afghanistan	1952	8425333.0	Asia	28.801	779.445314
99	Bangladesh	1967	62821884.0	Asia	43.453	721.186086
999	Mongolia	1967	1149500.0	Asia	51.253	1226.041130

Subset Rows by Row Number: iloc

- *# get the 2nd row*

```
In [41]: print(df.iloc[1])
```

country	Afghanistan
year	1957
pop	9.24093e+06
continent	Asia
lifeExp	30.332
gdpPercap	820.853

Name: 1, dtype: object

- *get the 100th row*

```
In [42]: print(df.iloc[99])
```

```
country      Bangladesh  
year         1967  
pop          6.28219e+07  
continent    Asia  
lifeExp      43.453  
gdpPercap    721.186  
Name: 99, dtype: object
```

- *# using -1 to get the last row*

```
In [43]: print(df.iloc[-1])
```

country	Zimbabwe
year	2007
pop	1.23111e+07
continent	Africa
lifeExp	43.487
gdpPercap	469.709
Name: 1703, dtype: object	

With `iloc`, we can pass in the `-1` to get the last row—something we couldn't do with `loc`.

- *# get the first, 100th, and 1000th rows*

```
In [44]: print(df.iloc[[0, 99, 999]])
```

	country	year	pop	continent	lifeExp	gdpPercap
0	Afghanistan	1952	8425333.0	Asia	28.801	779.445314
99	Bangladesh	1967	62821884.0	Asia	43.453	721.186086
999	Mongolia	1967	1149500.0	Asia	51.253	1226.041130

Subsetting Columns

- The Python slicing syntax uses a colon, :
- If we have just a colon, the attribute refers to everything.
- So, if we just want to get the first column using the loc or iloc syntax, we can write something like `df.loc[:, [columns]]` to subset the column(s).

- *# subset columns with loc*
note the position of the colon
it is used to select all rows

```
In [45]: subset = df.loc[:, ['year', 'pop']]  
print(subset.head())
```

	year	pop
0	1952	8425333.0
1	1957	9240934.0
2	1962	10267083.0
3	1967	11537966.0
4	1972	13079460.0

- # subset columns with iloc
- # iloc will allow us to use integers
- # -1 will select the last column

```
In [51]: subset = df.iloc[:, [2, 4, -1]]  
print(subset.head())
```

	pop	lifeExp	gdpPercap
0	8425333.0	28.801	779.445314
1	9240934.0	30.332	820.853030
2	10267083.0	31.997	853.100710
3	11537966.0	34.020	836.197138
4	13079460.0	36.088	739.981106

Subsetting Columns by Range

- *# create a range of integers from 0 to 4 inclusive*

```
In [52]: small_range = list(range(5))  
print(small_range)
```

```
[0, 1, 2, 3, 4]
```

- *# subset the dataframe with the range*

```
In [53]: subset = df.iloc[:, small_range]  
print(subset.head())
```

	country	year	pop	continent	lifeExp
0	Afghanistan	1952	8425333.0	Asia	28.801
1	Afghanistan	1957	9240934.0	Asia	30.332
2	Afghanistan	1962	10267083.0	Asia	31.997
3	Afghanistan	1967	11537966.0	Asia	34.020
4	Afghanistan	1972	13079460.0	Asia	36.088

Subsetting Rows and Columns

- *# using loc*

```
In [54]: # using loc  
print(df.loc[42, 'country'])
```

Angola

- *# using iloc*

```
In [55]: print(df.iloc[42, 0])
```

Angola

Subsetting Multiple Rows and Columns

- #get the 1st, 100th, and 1000th rows
#from the 1st, 4th, and 6th columns*

```
In [56]: print(df.iloc[[0, 99, 999], [0, 3, 5]])
```

	country	continent	gdpPercap
0	Afghanistan	Asia	779.445314
99	Bangladesh	Asia	721.186086
999	Mongolia	Asia	1226.041130

- *if we use the column names directly,
it makes the code a bit easier to read
note now we have to use loc, instead of iloc*

```
In [57]: print(df.loc[[0, 99, 999], ['country', 'lifeExp', 'gdpPercap']])
```

	country	lifeExp	gdpPercap
0	Afghanistan	28.801	779.445314
99	Bangladesh	43.453	721.186086
999	Mongolia	51.253	1226.041130

```
In [58]: print(df.loc[10:13, ['country', 'lifeExp', 'gdpPercap']])
```

	country	lifeExp	gdpPercap
10	Afghanistan	42.129	726.734055
11	Afghanistan	43.828	974.580338
12	Albania	55.230	1601.056136
13	Albania	59.280	1942.284244

```
In [59]: print(df.head(n=10))
```

	country	year	pop	continent	lifeExp	gdpPercap
0	Afghanistan	1952	8425333.0	Asia	28.801	779.445314
1	Afghanistan	1957	9240934.0	Asia	30.332	820.853030
2	Afghanistan	1962	10267083.0	Asia	31.997	853.100710
3	Afghanistan	1967	11537966.0	Asia	34.020	836.197138
4	Afghanistan	1972	13079460.0	Asia	36.088	739.981106
5	Afghanistan	1977	14880372.0	Asia	38.438	786.113360
6	Afghanistan	1982	12881816.0	Asia	39.854	978.011439
7	Afghanistan	1987	13867957.0	Asia	40.822	852.395945
8	Afghanistan	1992	16317921.0	Asia	41.674	649.341395
9	Afghanistan	1997	22227415.0	Asia	41.763	635.341351

Grouped Means

- *# For each year in our data, what was the average life expectancy?*
To answer this question,
we need to split our data into parts by year;
then we get the 'lifeExp' column and calculate the mean

```
In [60]: print(df.groupby('year')['lifeExp'].mean())
```

```
year
1952    49.057620
1957    51.507401
1962    53.609249
1967    55.678290
1972    57.647386
1977    59.570157
1982    61.533197
1987    63.212613
1992    64.160338
1997    65.014676
2002    65.694923
2007    67.007423
Name: lifeExp, dtype: float64
```

```
In [61]: multi_group_var = df.\
        groupby(['year', 'continent'])\
        [['lifeExp', 'gdpPercap']].\
        mean()\
print(multi_group_var)
```

		lifeExp	gdpPercap
year	continent		
1952	Africa	39.135500	1252.572466
	Americas	53.279840	4079.062552
	Asia	46.314394	5195.484004
	Europe	64.408500	5661.057435
	Oceania	69.255000	10298.085650
1957	Africa	41.266346	1385.236062
	Americas	55.960280	4616.043733
	Asia	49.318544	5787.732940
	Europe	66.703067	6963.012816
	Oceania	70.295000	11598.522455
1962	Africa	43.319442	1598.078825
	Americas	58.398760	4901.541870
	Asia	51.563772	5770.360675

- If you need to “flatten” the dataframe, you can use the `reset_index` method.

```
In [62]: flat = multi_group_var.reset_index()
print(flat.head(15))
```

	year	continent	lifeExp	gdpPercap
0	1952	Africa	39.135500	1252.572466
1	1952	Americas	53.279840	4079.062552
2	1952	Asia	46.314394	5195.484004
3	1952	Europe	64.408500	5661.057435
4	1952	Oceania	69.255000	10298.085650
5	1957	Africa	41.266346	1385.236062
6	1957	Americas	55.960280	4616.043733
7	1957	Asia	49.318544	5787.732940
8	1957	Europe	66.703067	6963.012816
9	1957	Oceania	70.295000	11598.522455
10	1962	Africa	43.319442	1598.078825
11	1962	Americas	58.398760	4901.541870
12	1962	Asia	51.563223	5729.369625
13	1962	Europe	68.539233	8365.486814
14	1962	Oceania	71.085000	12696.452430

Grouped Frequency Counts

- use the `nunique` to get counts of unique values on a Pandas Series.

```
In [63]: print(df.groupby('continent')['country'].nunique())
```

```
continent
Africa      52
Americas    25
Asia        33
Europe      30
Oceania      2
Name: country, dtype: int64
```

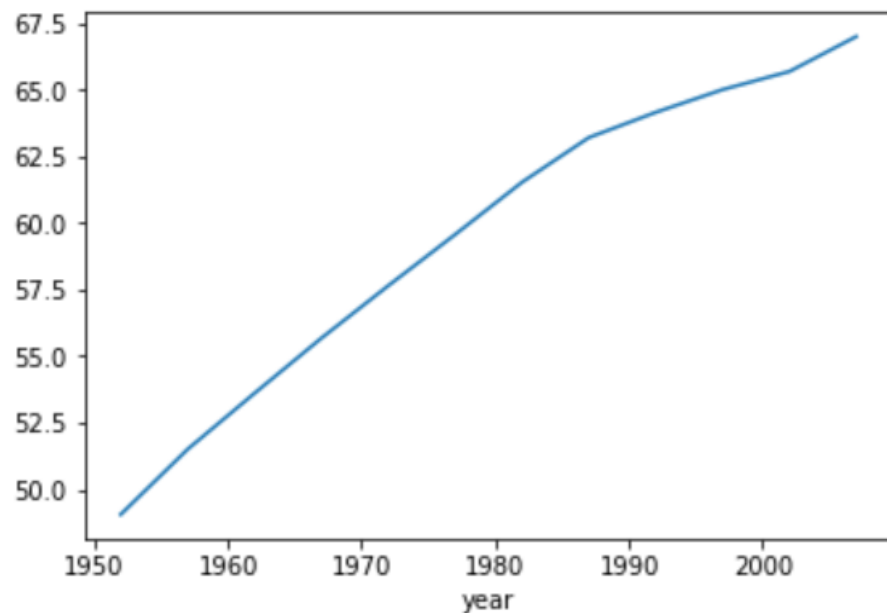
Basic Plot

```
In [65]: global_yearly_life_expectancy = df.groupby('year')['lifeExp'].mean()  
print(global_yearly_life_expectancy)
```

```
year  
1952    49.057620  
1957    51.507401  
1962    53.609249  
1967    55.678290  
1972    57.647386  
1977    59.570157  
1982    61.533197  
1987    63.212613  
1992    64.160338  
1997    65.014676  
2002    65.694923  
2007    67.007423  
Name: lifeExp, dtype: float64
```

```
In [66]: global_yearly_life_expectancy.plot()
```

```
Out[66]: <matplotlib.axes._subplots.AxesSubplot at 0x229d38dd320>
```



Visual Representation of the Data

- Histogram -- vertical bar chart of frequencies
- Frequency Polygon -- line graph of frequencies
- Ogive -- line graph of cumulative frequencies
- Pie Chart -- proportional representation for categories of a whole
- Stem and Leaf Plot
- Pareto Chart
- Scatter Plot

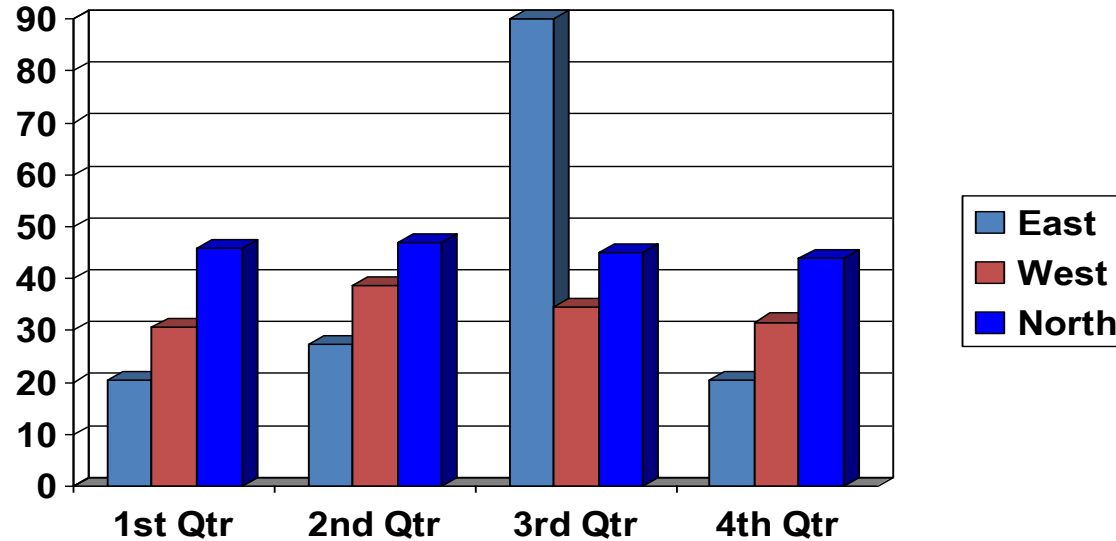
Methods of visual presentation of data

- Table

	1st Qtr	2nd Qtr	3rd Qtr	4th Qtr
East	20.4	27.4	90	20.4
West	30.6	38.6	34.6	31.6
North	45.9	46.9	45	43.9

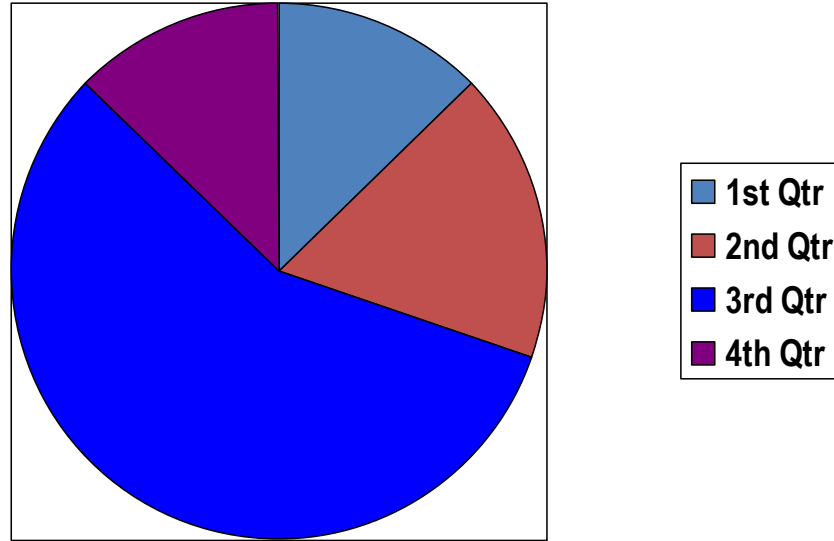
Methods of visual presentation of data

- Graphs



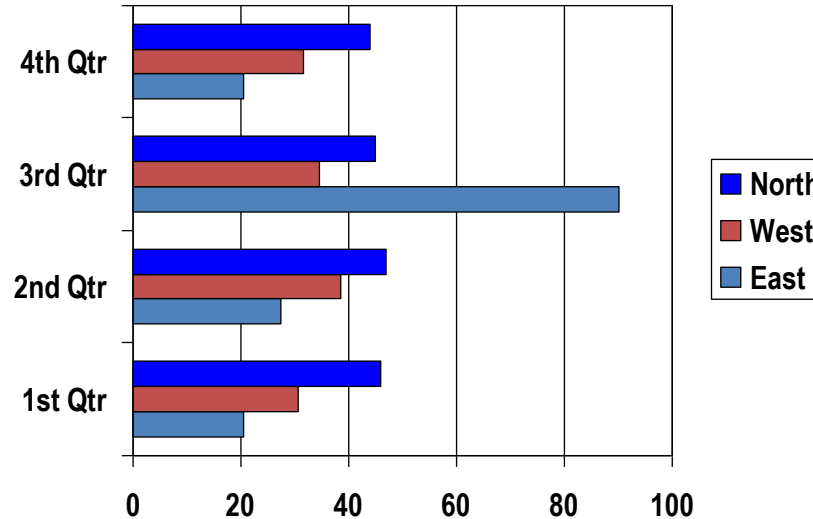
Methods of visual presentation of data

- Pie chart



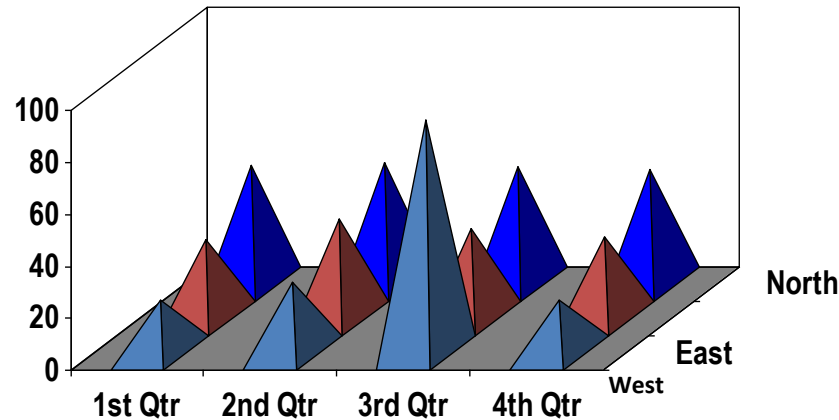
Methods of visual presentation of data

- Multiple bar chart



Methods of visual presentation of data

- Simple pictogram

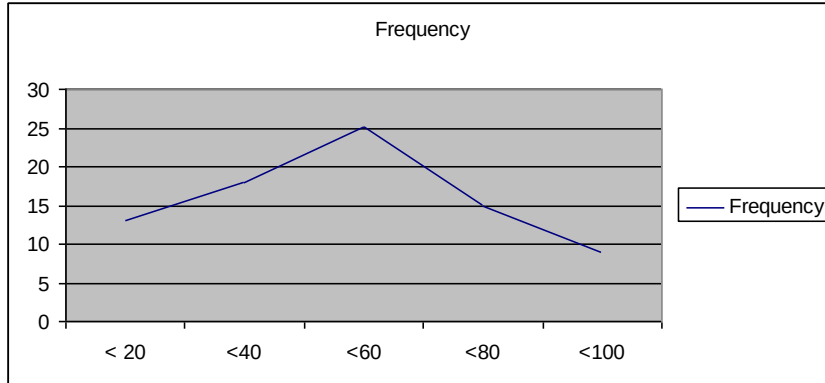
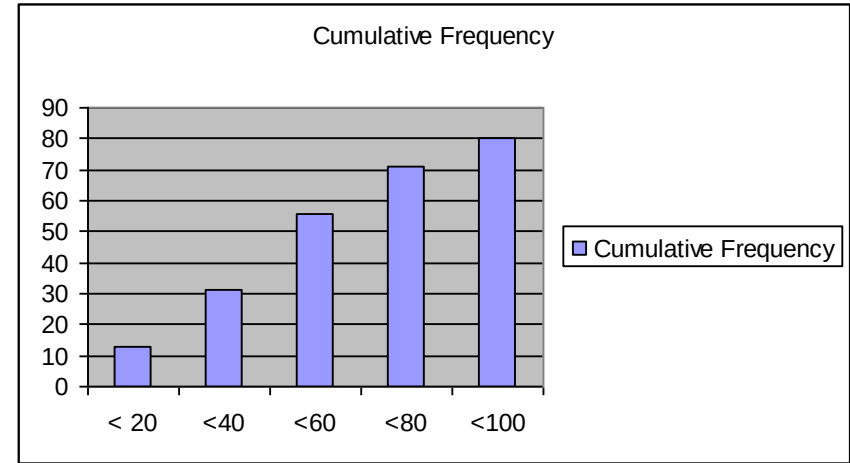
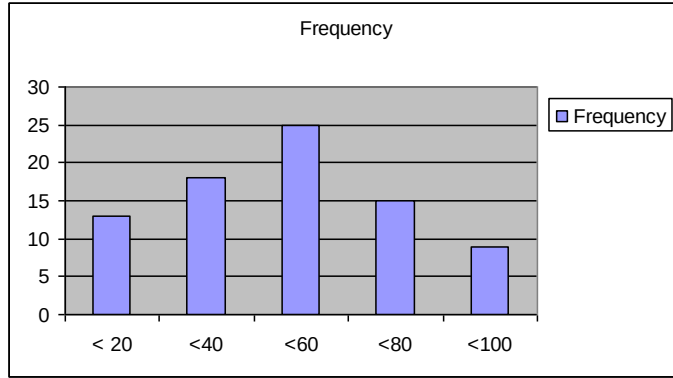


Frequency distributions

- Frequency tables

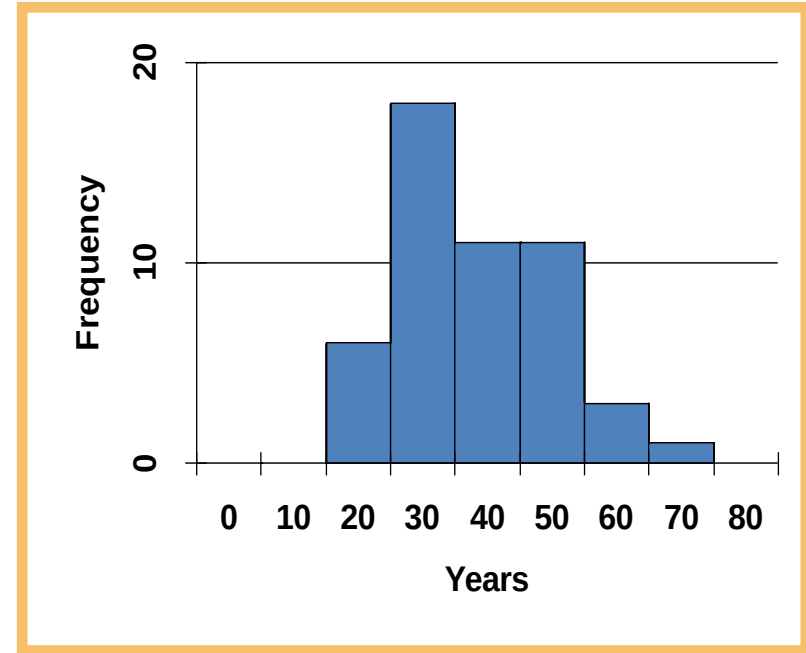
Observation Table		
Class Interval	Frequency	Cumulative Frequency
< 20	13	13
<40	18	31
<60	25	56
<80	15	71
<100	9	80

Frequency diagrams



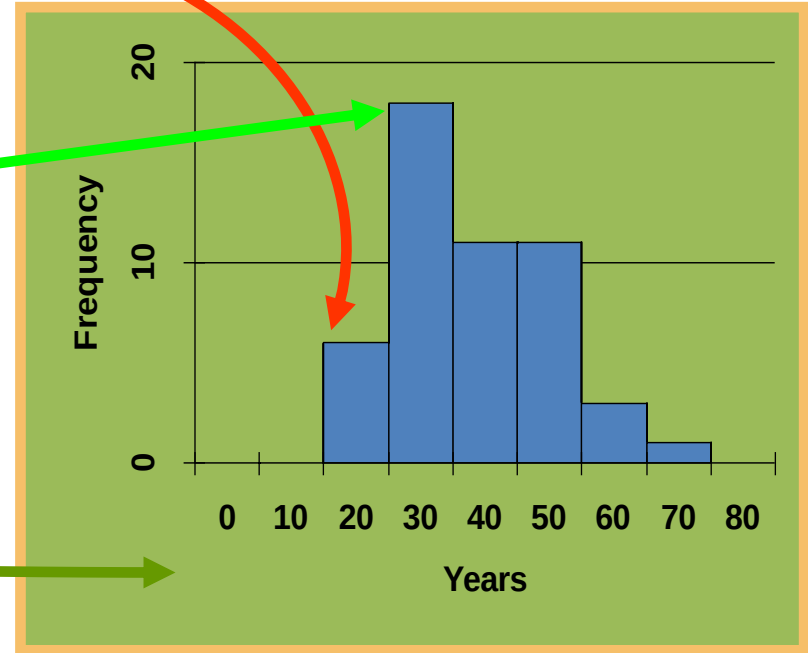
Histogram

<u>Class Interval</u>	<u>Frequency</u>
20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1



Histogram Construction

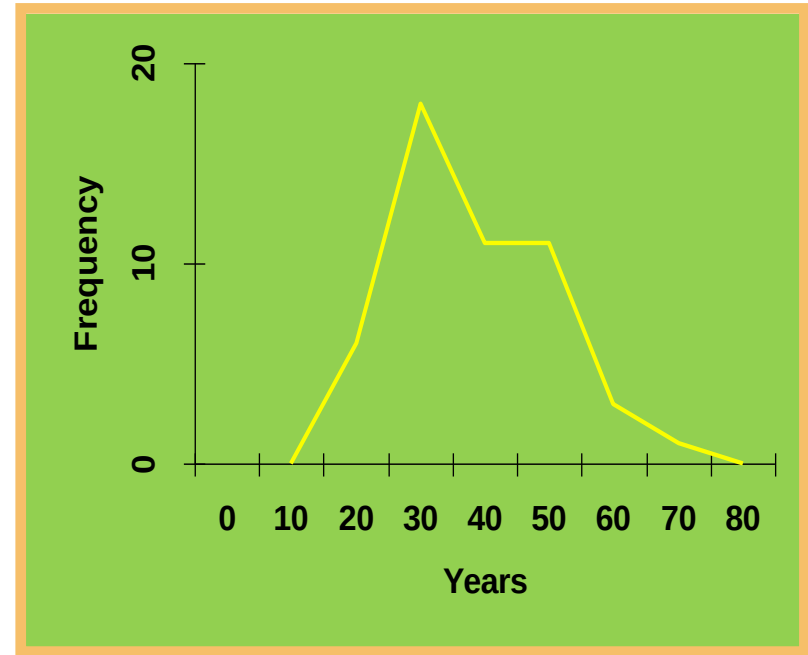
Class Interval	Frequency
20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1



Frequency Polygon

Class IntervalFrequency

20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1



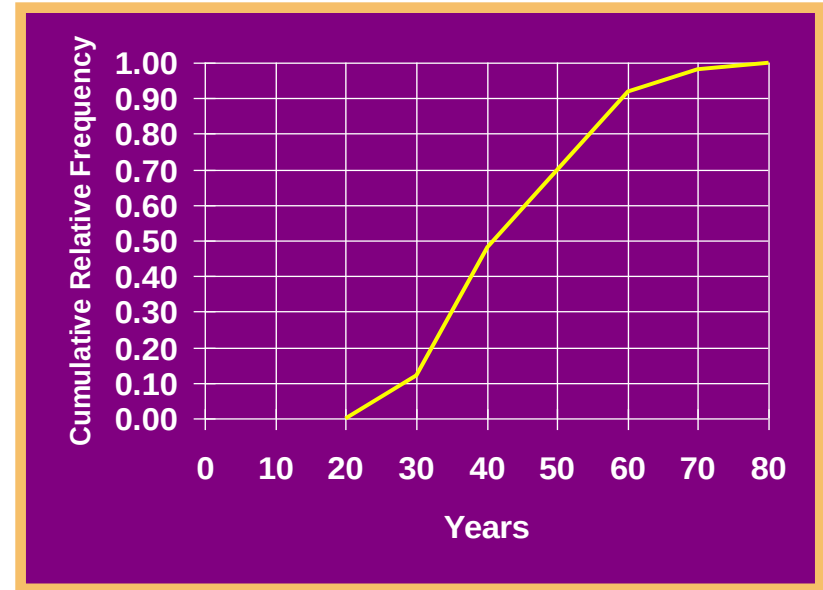
Ogive

<u>Class Interval</u>	<u>Cumulative Frequency</u>
20-under 30	6
30-under 40	24
40-under 50	35
50-under 60	46
60-under 70	49
70-under 80	50

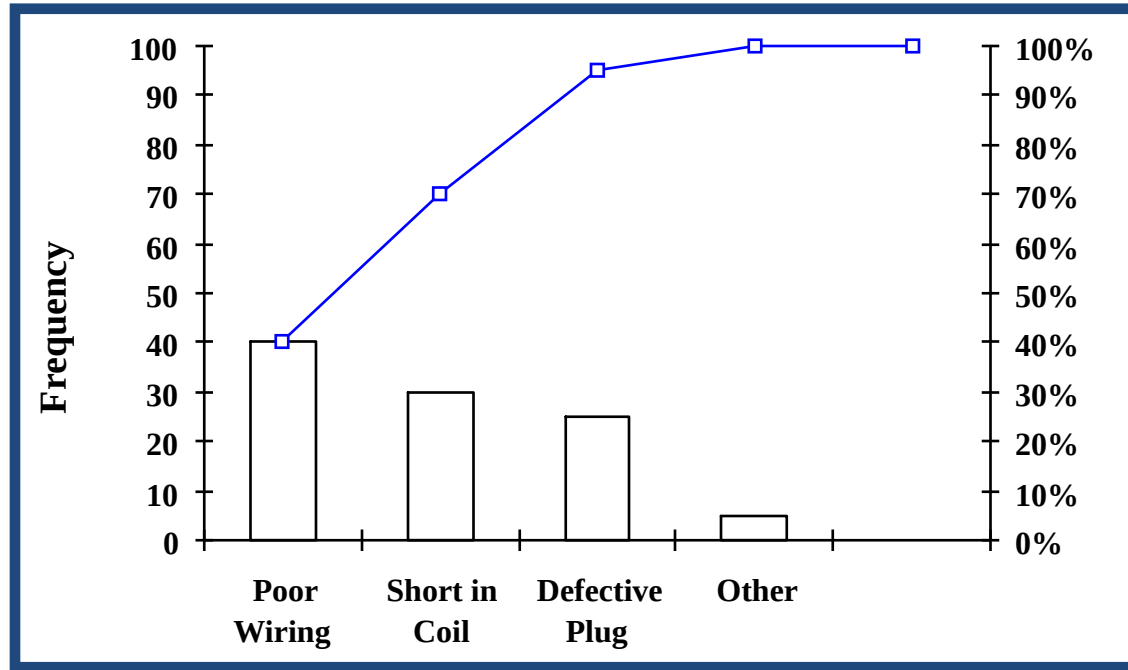


Relative Frequency Ogive

<u>Class Interval</u>	<u>Cumulative Relative Frequency</u>
20-under 30	.12
30-under 40	.48
40-under 50	.70
50-under 60	.92
60-under 70	.98
70-under 80	1.00



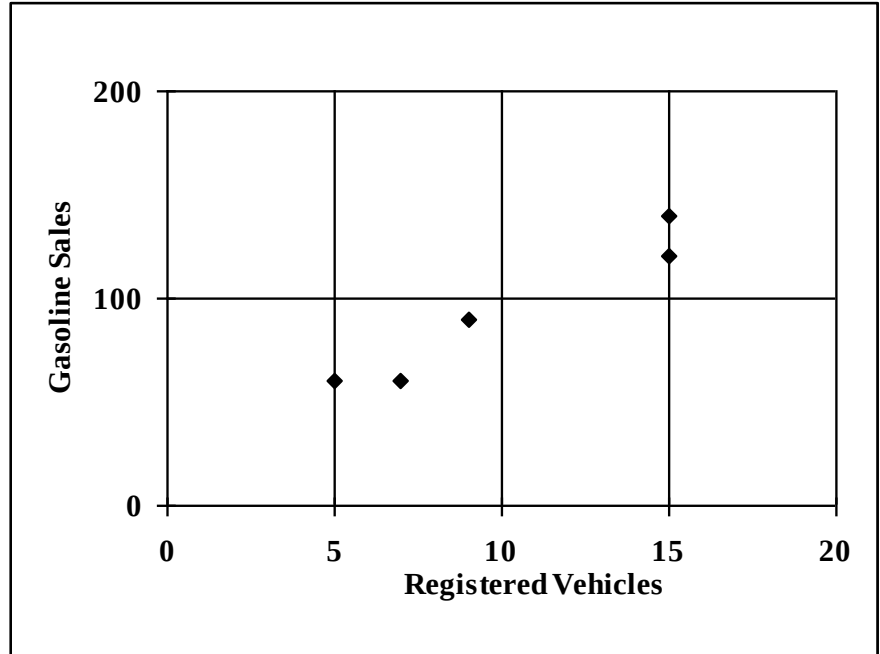
Pareto Chart



Scatter Plot

Registered Vehicles (1000's)	Gasoline Sales (1000's of Gallons)
---------------------------------	---------------------------------------

5	60
15	120
9	90
15	140
7	60



Principles of Excellent Graphs

- The graph should not distort the data
- The graph should not contain unnecessary adornments (sometimes referred to as chart junk)
- The scale on the vertical axis should begin at zero
- All axes should be properly labeled
- The graph should contain a title
- The simplest possible graph should be used for a given set of data

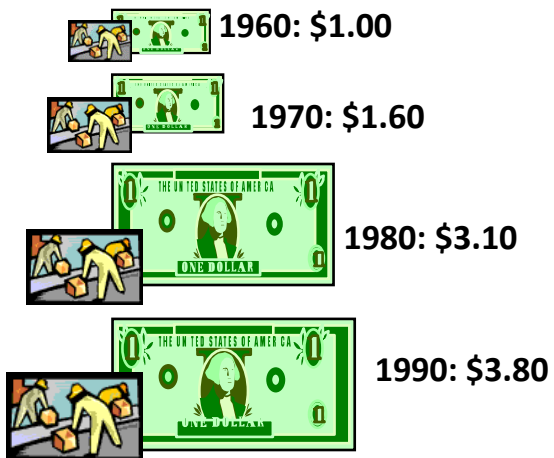


Graphical Errors: Chart Junk



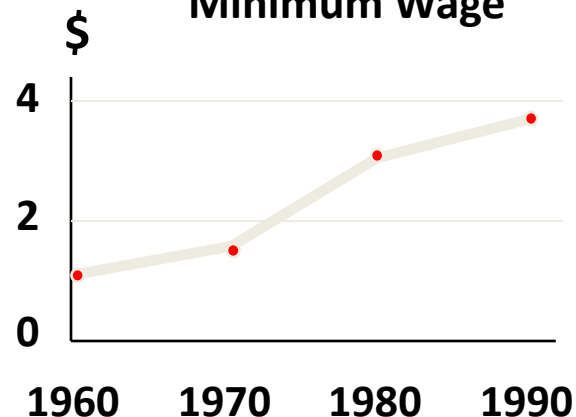
Bad Presentation

Minimum Wage



Good Presentation

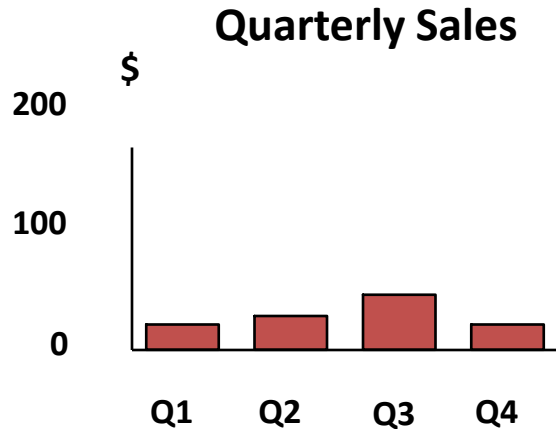
Minimum Wage



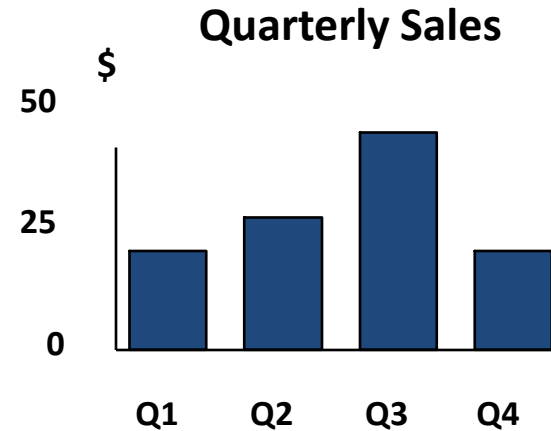
Graphical Errors: Compressing the Vertical Axis



Bad Presentation



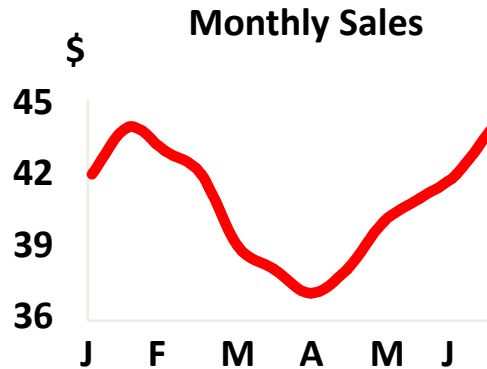
Good Presentation



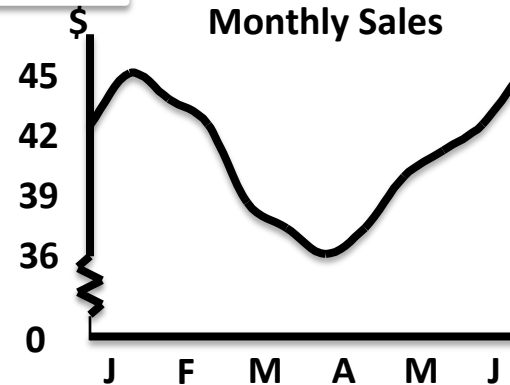
Graphical Errors: No Zero Point on the Vertical Axis



Bad Presentation



Good Presentations



Graphing the first six months of sales



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Lecture 4: Central Tendency and Dispersion

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Department of Management Studies



Lecture objectives

- Central tendency
- Measures of Dispersion

Measures of Central Tendency

- Measures of central tendency yield information about “particular places or locations in a group of numbers.”
- A single number to describe the characteristics of a set of data

Summary statistics

- Central tendency or measures of location
 - Arithmetic mean
 - Weighted mean
 - Median
 - Percentile
- Dispersion
 - Skewness
 - Kurtosis
 - Range
 - Interquartile range
 - Variance
 - Standard score
 - Coefficient of variation

Arithmetic Mean

- Commonly called 'the mean'
- It is the average of a group of numbers
- Applicable for interval and ratio data
- Not applicable for nominal or ordinal data
- Affected by each value in the data set, including extreme values
- Computed by summing all values in the data set and dividing the sum by the number of values in the data set

Population Mean

$$\begin{aligned}\mu &= \frac{\sum X}{N} = \frac{X_1 + X_2 + X_3 + \dots + X_N}{N} \\ &= \frac{24 + 13 + 19 + 26 + 11}{5} \\ &= \frac{93}{5} \\ &= 18.6\end{aligned}$$

Sample Mean

$$\begin{aligned}\bar{X} &= \frac{\sum X}{n} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \\ &= \frac{57 + 86 + 42 + 38 + 90 + 66}{6} \\ &= \frac{379}{6} \\ &= 63.167\end{aligned}$$

Mean of Grouped Data

- Weighted average of class midpoints
- Class frequencies are the weights

$$\begin{aligned}\mu &= \frac{\sum fM}{\sum f} \\ &= \frac{\sum fM}{N} \\ &= \frac{f_1M_1 + f_2M_2 + f_3M_3 + \cdots + f_iM_i}{f_1 + f_2 + f_3 + \cdots + f_i}\end{aligned}$$

Calculation of Grouped Mean

Class Interval	Frequency(f)	Class Midpoint(M)	fM
20-under 30	6	25	150
30-under 40	18	35	630
40-under 50	11	45	495
50-under 60	11	55	605
60-under 70	3	65	195
70-under 80	<u>1</u>	75	<u>75</u>
	50		2150

$$\mu = \frac{\sum fM}{\sum f} = \frac{2150}{50} = 43.0$$

Weighted Average

- Sometimes we wish to average numbers, but we want to assign more importance, or weight, to some of the numbers.
- The average you need is the *weighted average*.



Formula for Weighted Average

$$\text{Weighted Average} = \frac{\sum xw}{\sum w}$$

where x is a data value and w is the weight assigned to that data value. The sum is taken over all data values.

Example

Suppose your midterm test score is 83 and your final exam score is 95. Using weights of 40% for the midterm and 60% for the final exam, compute the weighted average of your scores. If the minimum average for an A is 90, will you earn an A?

$$\begin{aligned}\text{Weighted Average} &= \frac{(83)(0.40) + (95)(0.60)}{0.40 + 0.60} \\ &= \frac{32 + 57}{1} = 90.2\end{aligned}$$

You will earn an A!

Median

- Middle value in an ordered array of numbers
- Applicable for ordinal, interval, and ratio data
- Not applicable for nominal data
- Unaffected by extremely large and extremely small values

Median: Computational Procedure

- First Procedure
 - Arrange the observations in an ordered array
 - If there is an odd number of terms, the median is the middle term of the ordered array
 - If there is an even number of terms, the median is the average of the middle two terms
- Second Procedure
 - The median's position in an ordered array is given by $(n+1)/2$.

Median: Example with an Odd Number of Terms

Ordered Array

3 4 5 7 8 9 11 14 15 16 16 17 19 19 20 21 22

- There are 17 terms in the ordered array.
- Position of median = $(n+1)/2 = (17+1)/2 = 9$
- The median is the 9th term, 15.
- If the 22 is replaced by 100, the median is 15.
- If the 3 is replaced by -103, the median is 15.

Median: Example with an Even Number of Terms

Ordered Array

3 4 5 7 8 9 11 14 15 16 16 17 19 19 20 21

- There are 16 terms in the ordered array
- Position of median = $(n+1)/2 = (16+1)/2 = 8.5$
- The median is between the 8th and 9th terms, 14.5
- If the 21 is replaced by 100, the median is 14.5
- If the 3 is replaced by -88, the median is 14.5

Median of Grouped Data

$$\text{Median} = L + \frac{\frac{N}{2} - cf_p}{f_{med}} (W)$$

Where :

L = the lower limit of the median class

cf_p = cumulative frequency of class preceding the median class

f_{med} = frequency of the median class

W = width of the median class

N = total of frequencies

Median of Grouped Data -- Example

<u>Class Interval</u>	<u>Frequency</u>	<u>Cumulative Frequency</u>
20-under 30	6	6
30-under 40	18	24
40-under 50	11	35
50-under 60	11	46
60-under 70	3	49
70-under 80	<u>1</u>	50
N = 50		

$$\begin{aligned}Md &= L + \frac{\frac{N}{2} - cf_p}{f_{med}}(W) \\&= 40 + \frac{\frac{50}{2} - 24}{11}(10) \\&= 40.909\end{aligned}$$

Mode

- The most frequently occurring value in a data set
- Applicable to all levels of data measurement (nominal, ordinal, interval, and ratio)
- Bimodal -- Data sets that have two modes
- Multimodal -- Data sets that contain more than two modes

Mode -- Example

- The mode is 44
- There are more 44s than any other value

35	41	44	45
37	41	44	46
37	43	44	46
39	43	44	46
40	43	44	46
40	43	45	48

Mode of Grouped Data

- Midpoint of the modal class
- Modal class has the greatest frequency

<u>Class Interval</u>	<u>Frequency</u>
20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1

$$Mode = L_{Mo} + \left(\frac{d_1}{d_1 + d_2} \right) w =$$

$$30 + \left(\frac{12}{12 + 7} \right) 10 = 36.31$$



Percentiles

- Measures of central tendency that divide a group of data into 100 parts
- Example: 90th percentile indicates that at most 90% of the data lie below it, and at least 10% of the data lie above it
- The median and the 50th percentile have the same value
- Applicable for ordinal, interval, and ratio data
- Not applicable for nominal data

Percentiles: Computational Procedure

- Organize the data into an ascending ordered array
- Calculate the p th percentile location:

$$i = \frac{P}{100}(n)$$

- Determine the percentile's location and its value.
- If i is a **whole number**, the percentile is the average of the values at the i and $(i+1)$ positions
- If i is **not a whole number**, the percentile is at the $(i+1)$ position in the ordered array

Percentiles: Example

- Raw Data: 14, 12, 19, 23, 5, 13, 28, 17
- Ordered Array: 5, 12, 13, 14, 17, 19, 23, 28
- Location of 30th percentile:

$$i = \frac{30}{100}(8) = 2.4$$

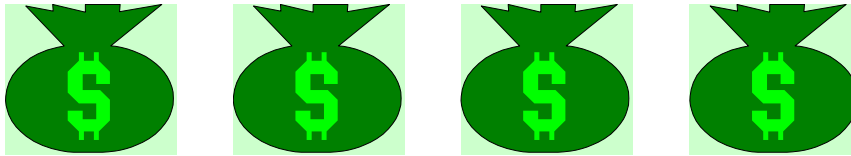
- The location index, i , is not a whole number; $i+1 = 2.4+1=3.4$; the whole number portion is 3; the 30th percentile is at the 3rd location of the array; the 30th percentile is 13.

Dispersion

- Measures of variability describe the spread or the dispersion of a set of data
- Reliability of measure of central tendency
- To compare dispersion of various samples

Variability

No Variability in Cash Flow



Mean



Variability in Cash Flow



Mean



Measures of Variability or dispersion

Common Measures of Variability

- Range
- Inter-quartile range
- Mean Absolute Deviation
- Variance
- Standard Deviation
- Z scores
- Coefficient of Variation

Range – ungrouped data

- The difference between the largest and the smallest values in a set of data
- Simple to compute
- Ignores all data points except the two extremes
- Example:

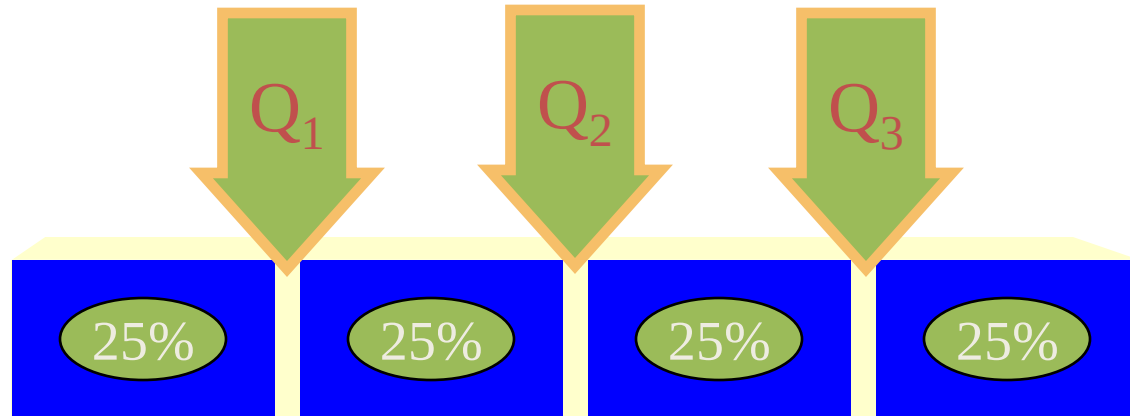
$$\text{Range} = \text{Largest} - \text{Smallest} = 48 - 35 = 13$$

35	41	44	45
37	41	44	46
37	43	44	46
39	43	44	46
40	43	44	46
40	43	45	48

Quartiles

- Measures of central tendency that divide a group of data into four subgroups
- Q_1 : 25% of the data set is below the first quartile
- Q_2 : 50% of the data set is below the second quartile
- Q_3 : 75% of the data set is below the third quartile
- Q_1 is equal to the 25th percentile
- Q_2 is located at 50th percentile and equals the median
- Q_3 is equal to the 75th percentile
- Quartile values are not necessarily members of the data set

Quartiles



Quartiles: Example

- Ordered array: 106, 109, 114, 116, 121, 122, 125, 129

- Q_1 $i = \frac{25}{100}(8) = 2$ $Q_1 = \frac{109+114}{2} = 111.5$

- Q_2 : $i = \frac{50}{100}(8) = 4$ $Q_2 = \frac{116+121}{2} = 118.5$

- Q_3 : $i = \frac{75}{100}(8) = 6$ $Q_3 = \frac{122+125}{2} = 123.5$

Interquartile Range

- Range of values between the first and third quartiles
- Range of the “middle half”
- Less influenced by extremes

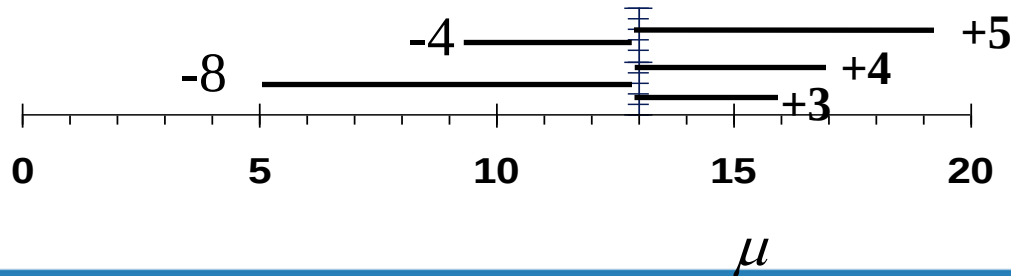
$$\text{Interquartile Range} = Q_3 - Q_1$$

Deviation from the Mean

- Data set: 5, 9, 16, 17, 18

- Mean: $\mu = \frac{\sum X}{N} = \frac{65}{5} = 13$

- Deviations from the mean: -8, -4, 3, 4, 5



Mean Absolute Deviation

- Average of the absolute deviations from the mean

X	$X - \mu$	$ X - \mu $
5	-8	+8
9	-4	+4
16	+3	+3
17	+4	+4
18	<u>+5</u>	<u>+5</u>
	0	24

$$\begin{aligned} M.A.D. &= \frac{\sum |X - \mu|}{N} \\ &= \frac{24}{5} \\ &= 4.8 \end{aligned}$$

Population Variance

- Average of the squared deviations from the arithmetic mean

X	$X - \mu$	$(X - \mu)^2$
5	-8	64
9	-4	16
16	+3	9
17	+4	16
18	+5	25
	<u>0</u>	<u>130</u>

$$\begin{aligned}\sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\ &= \frac{130}{5} \\ &= 26.0\end{aligned}$$

Population Standard Deviation

- Square root of the variance

X	$X - \mu$	$(X - \mu)^2$
5	-8	64
9	-4	16
16	+3	9
17	+4	16
18	+5	25
	0	130

$$\begin{aligned}\sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\ &= \frac{130}{5} \\ &= 26.0 \\ \sigma &= \sqrt{\sigma^2} \\ &= \sqrt{26.0} \\ &= 5.1\end{aligned}$$

Sample Variance

- Average of the squared deviations from the arithmetic mean

X	$X - \bar{X}$	$(X - \bar{X})^2$
2,398	625	390,625
1,844	71	5,041
1,539	-234	54,756
<u>1,311</u>	<u>-462</u>	<u>213,444</u>
7,092	0	663,866

$$\begin{aligned} S^2 &= \frac{\sum (X - \bar{X})^2}{n-1} \\ &= \frac{663,866}{3} \\ &= 221,288.67 \end{aligned}$$

Sample Standard Deviation

- Square root of the sample variance

X	$X - \bar{X}$	$(X - \bar{X})^2$
2,398	625	390,625
1,844	71	5,041
1,539	-234	54,756
<u>1,311</u>	<u>-462</u>	<u>213,444</u>
7,092	0	663,866

$$\begin{aligned}
 S^2 &= \frac{\sum (X - \bar{X})^2}{n - 1} \\
 &= \frac{663,866}{3} \\
 &= 221,288.67 \\
 S &= \sqrt{S^2} \\
 &= \sqrt{221,288.67} \\
 &= 470.41
 \end{aligned}$$

Uses of Standard Deviation

- Indicator of financial risk
- Quality Control
 - construction of quality control charts
 - process capability studies
- Comparing populations
 - household incomes in two cities
 - employee absenteeism at two plants

Standard Deviation as an Indicator of Financial Risk

Financial Security	Annualized Rate of Return	
	μ	σ
A	15%	3%
B	15%	7%



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Lecture 5: Central Tendency and Dispersion- II

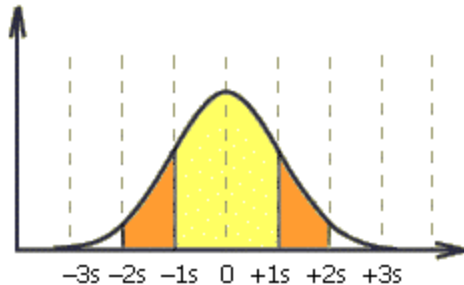
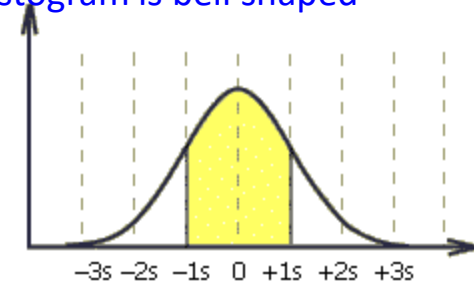
Dr. A. Ramesh

Department of Management Studies



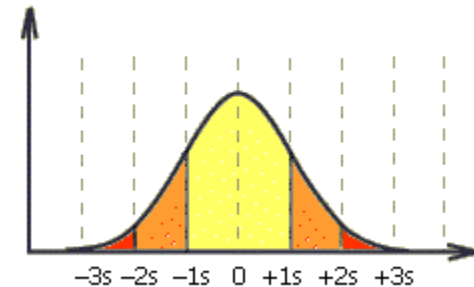
The Empirical Rule... If the histogram is bell shaped

- Approximately 68% of all observations fall within **one** standard deviation of the mean.



- Approximately 95% of all observations fall within **two** standard deviations of the mean.

- Approximately 99.7% of all observations fall within **three** standard deviations of the mean.



Empirical Rule

- Data are normally distributed (or approximately normal)

Distance from the Mean	Percentage of Values Falling Within Distance
$\mu \pm 1 \sigma$	68
$\mu \pm 2 \sigma$	95
$\mu \pm 3 \sigma$	99.7

Chebysheff's Theorem...Not often used because interval is very wide.

- A more general interpretation of the standard deviation is derived from ***Chebysheff's Theorem***, which applies to all shapes of histograms (not just bell shaped).
- The proportion of observations in any sample that lie within **k** standard deviations of the mean is *at least*:

$$1 - \frac{1}{k^2} \text{ for } k > 1$$

For $k=2$ (say), the theorem states that at least $3/4$ of all observations lie within 2 standard deviations of the mean. This is a "lower bound" compared to Empirical Rule's approximation (95%).

Coefficient of Variation

- Ratio of the standard deviation to the mean, expressed as a percentage
- Measurement of relative dispersion

$$C.V. = \frac{\sigma}{\mu}(100)$$

Coefficient of Variation

$$\mu_1 = 29$$

$$\sigma_1 = 4.6$$

$$\begin{aligned} C.V._1 &= \frac{\sigma_1}{\mu_1}(100) \\ &= \frac{4.6}{29}(100) \\ &= 15.86 \end{aligned}$$

$$\mu_2 = 84$$

$$\sigma_2 = 10$$

$$\begin{aligned} C.V._2 &= \frac{\sigma_2}{\mu_2}(100) \\ &= \frac{10}{84}(100) \\ &= 11.90 \end{aligned}$$

Variance and Standard Deviation of Grouped Data

Population

$$\sigma^2 = \frac{\sum f (M - \mu)^2}{N}$$
$$\sigma = \sqrt{\sigma^2}$$

Sample

$$S^2 = \frac{\sum f (M - \bar{X})^2}{n - 1}$$
$$S = \sqrt{S^2}$$

Population Variance and Standard Deviation of Grouped Data($\mu=43$)

<i>Class Interval</i>	f	M	fM	$M-\mu$	$(M-\mu)^2$	$f(M-\mu)^2$
20-under 30	6	25	150	-18	324	1944
30-under 40	18	35	630	-8	64	1152
40-under 50	11	45	495	2	4	44
50-under 60	11	55	605	12	144	1584
60-under 70	3	65	195	22	484	1452
70-under 80	1	75	<u>75</u>	32	1024	<u>1024</u>
	50		2150			7200

$$\sigma^2 = \frac{\sum f(M-\mu)^2}{N} = \frac{7200}{50} = 144$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{144} = 12$$

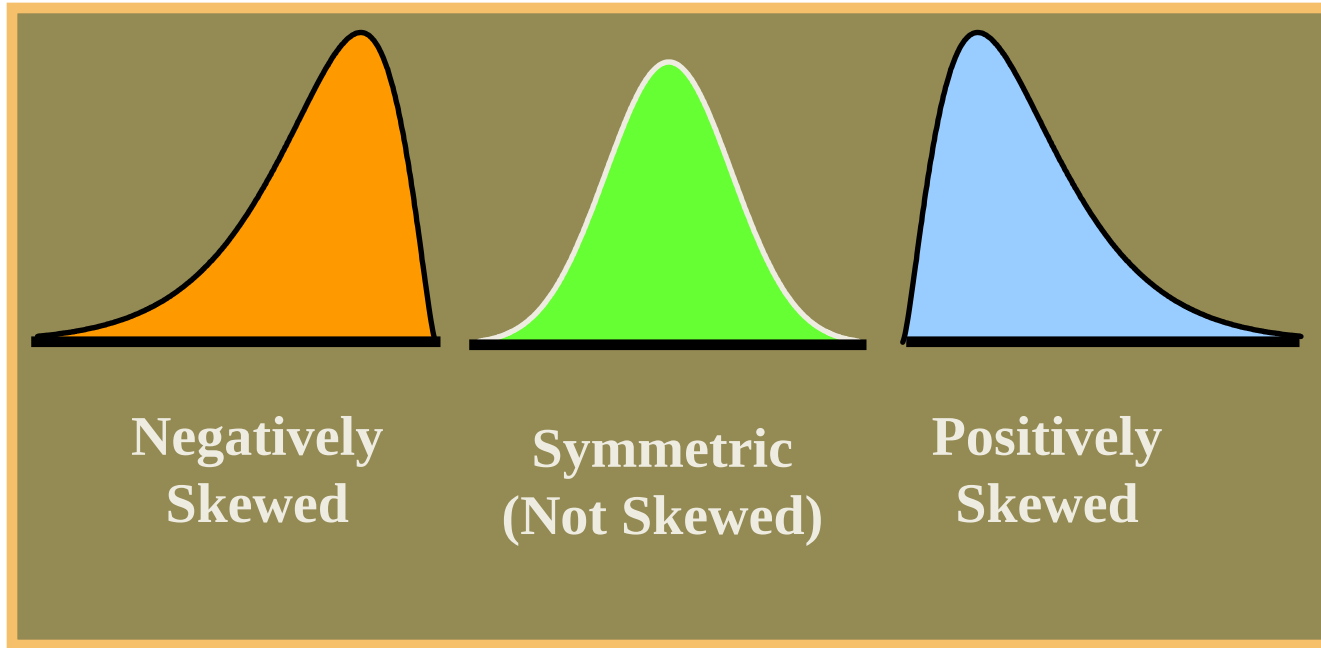
Measures of Shape

- **Skewness**
 - Absence of symmetry
 - Extreme values in one side of a distribution
- **Kurtosis**

Peakedness of a distribution

 - Leptokurtic: high and thin
 - Mesokurtic: normal shape
 - Platykurtic: flat and spread out
- **Box and Whisker Plots**
 - Graphic display of a distribution
 - Reveals skewness

Skewness

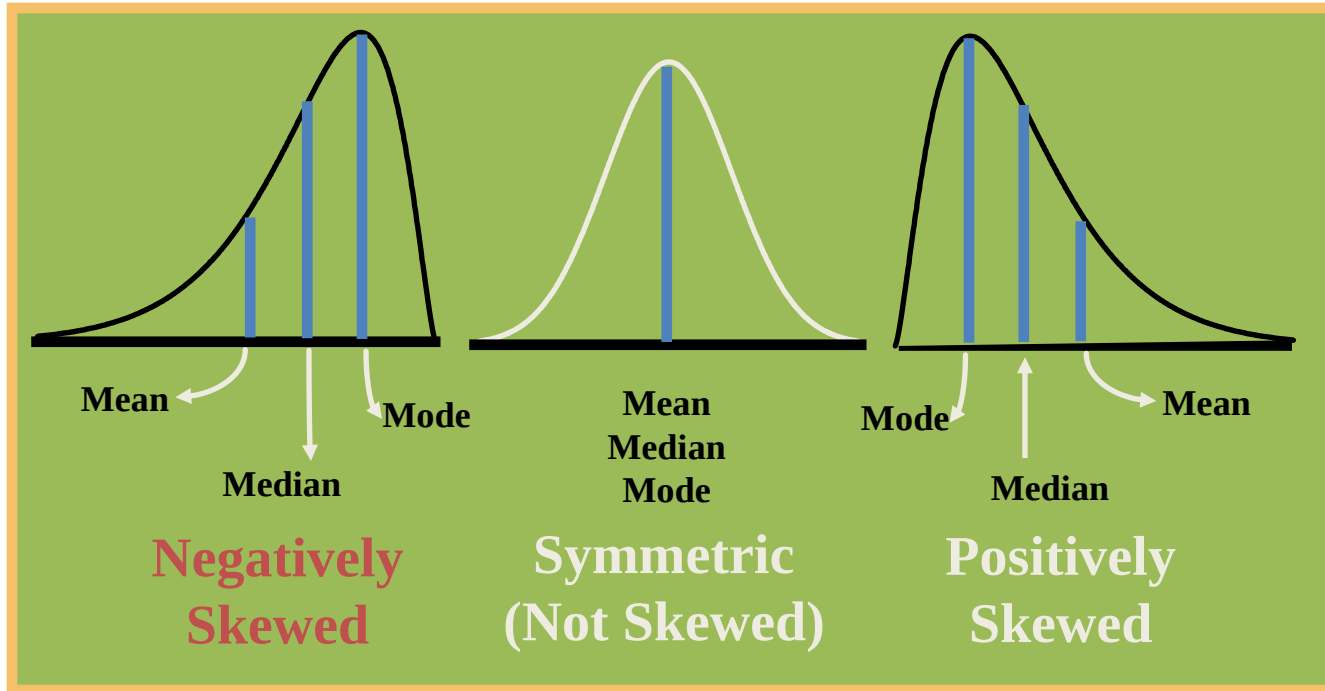


Skewness..

The *skewness* of a distribution is measured by comparing the relative positions of the mean, median and mode.

- Distribution is *symmetrical*
 - ***Mean = Median = Mode***
- Distribution *skewed right*
 - ***Median lies between mode and mean, and mode is less than mean***
- Distribution *skewed left*
 - ***Median lies between mode and mean, and mode is greater than mean***

Skewness



Coefficient of Skewness

- Summary measure for skewness

$$S = \frac{3(\mu - M_d)}{\sigma}$$

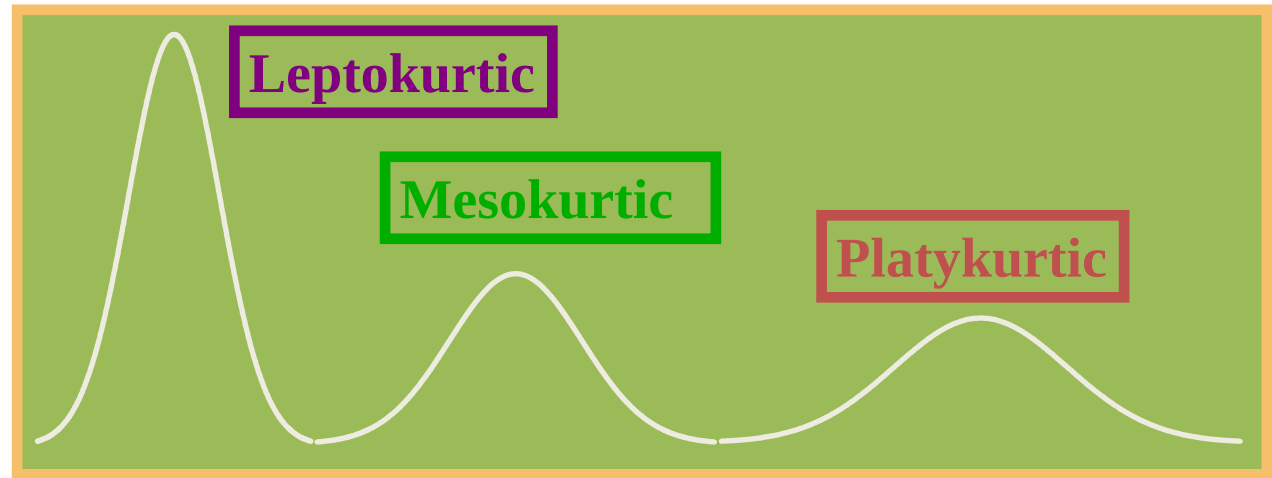
- If $S < 0$, the distribution is negatively skewed (skewed to the left)
- If $S = 0$, the distribution is symmetric (not skewed)
- If $S > 0$, the distribution is positively skewed (skewed to the right)

Coefficient of Skewness

$\mu_1 = 23$ $M_{d1} = 26$ $\sigma_1 = 12.3$ $S_1 = \frac{3(\mu_1 - M_{d1})}{\sigma_1}$ $= \frac{3(23 - 26)}{12.3}$ $= -0.73$	$\mu_2 = 26$ $M_{d2} = 26$ $\sigma_2 = 12.3$ $S_2 = \frac{3(\mu_2 - M_{d2})}{\sigma_2}$ $= \frac{3(26 - 26)}{12.3}$ $= 0$	$\mu_3 = 29$ $M_{d3} = 26$ $\sigma_3 = 12.3$ $S_3 = \frac{3(\mu_3 - M_{d3})}{\sigma_3}$ $= \frac{3(29 - 26)}{12.3}$ $= +0.73$
--	--	--

Kurtosis

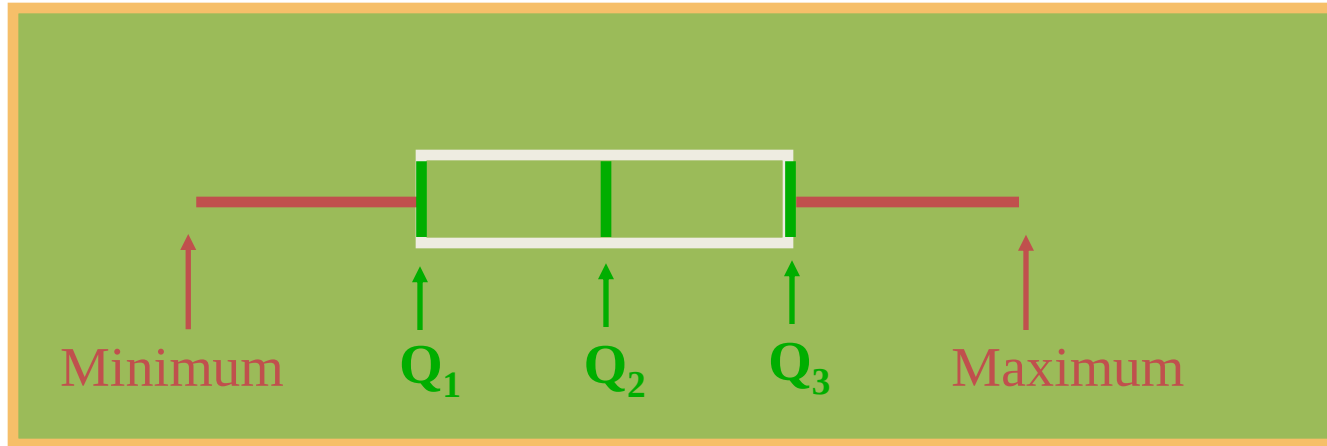
- Peakedness of a distribution
 - Leptokurtic: high and thin
 - Mesokurtic: normal in shape
 - Platykurtic: flat and spread out



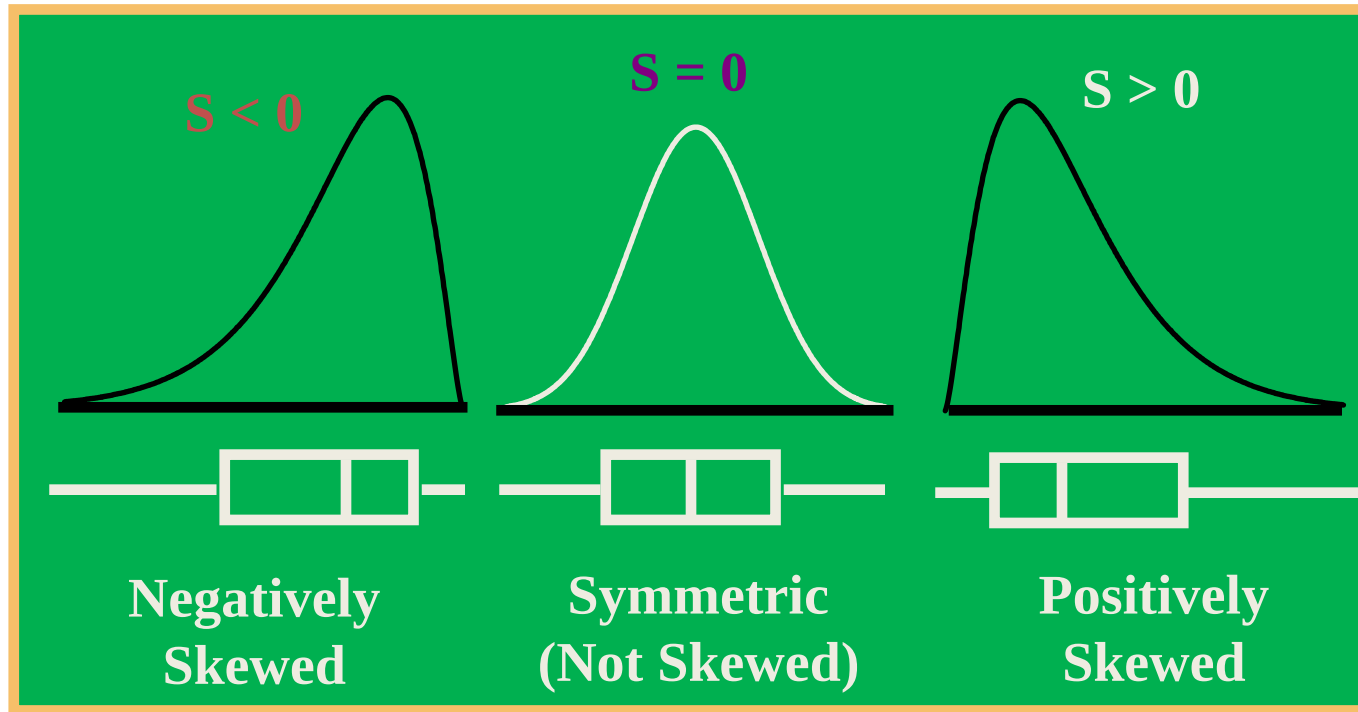
Box and Whisker Plot

- Five specific values are used:
 - Median, Q_2
 - First quartile, Q_1
 - Third quartile, Q_3
 - Minimum value in the data set
 - Maximum value in the data set

Box and Whisker Plot



Skewness: Box and Whisker Plots, and Coefficient of Skewness



THANK YOU

